AN AGENT-BASED SIMULATION FRAMEWORK FOR ONLINE OPTIMIZATION OF THE ON-DEMAND TRANSIT OPERATION

Defense Examination (M.Sc. Engg.)

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OUTLINE

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1. Introduction
2. Literature Review
3. Objectives
4. Outcomes
5. Method
6. Result Analysis
7. Conclusion
8. Future work

INTRODUCTION

Private car

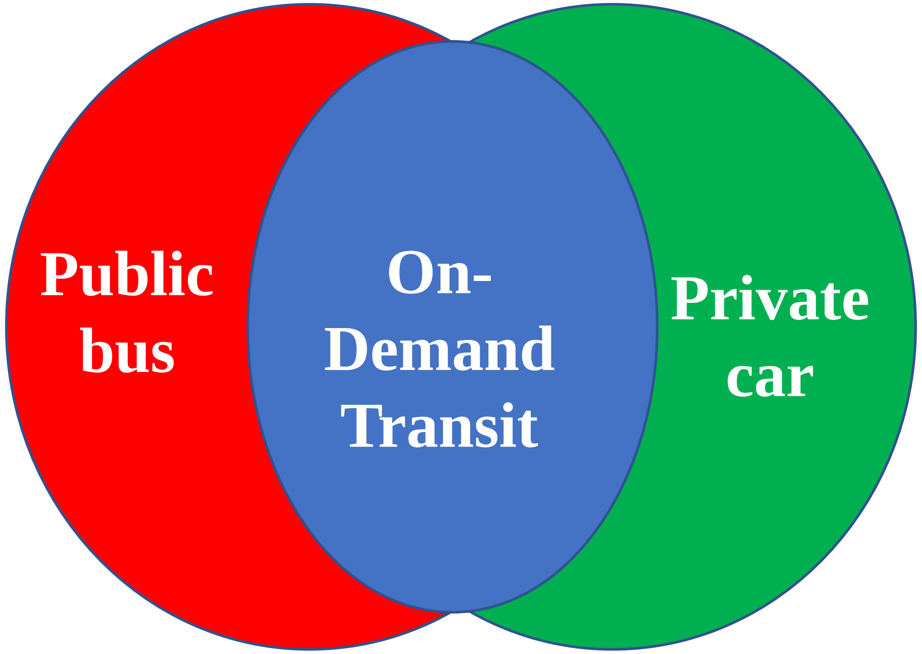
- Create traffic congestions in the road
- More air pollution
- More fuel consumption

Public bus

- Not available all time
- Fixed bus stations

INTRODUCTION

What if we could combine the best of both?



Flexibility in routing and scheduling

Routes can be updated before completing tour for previous route

Private car

Sharing of resources among multiple travelers

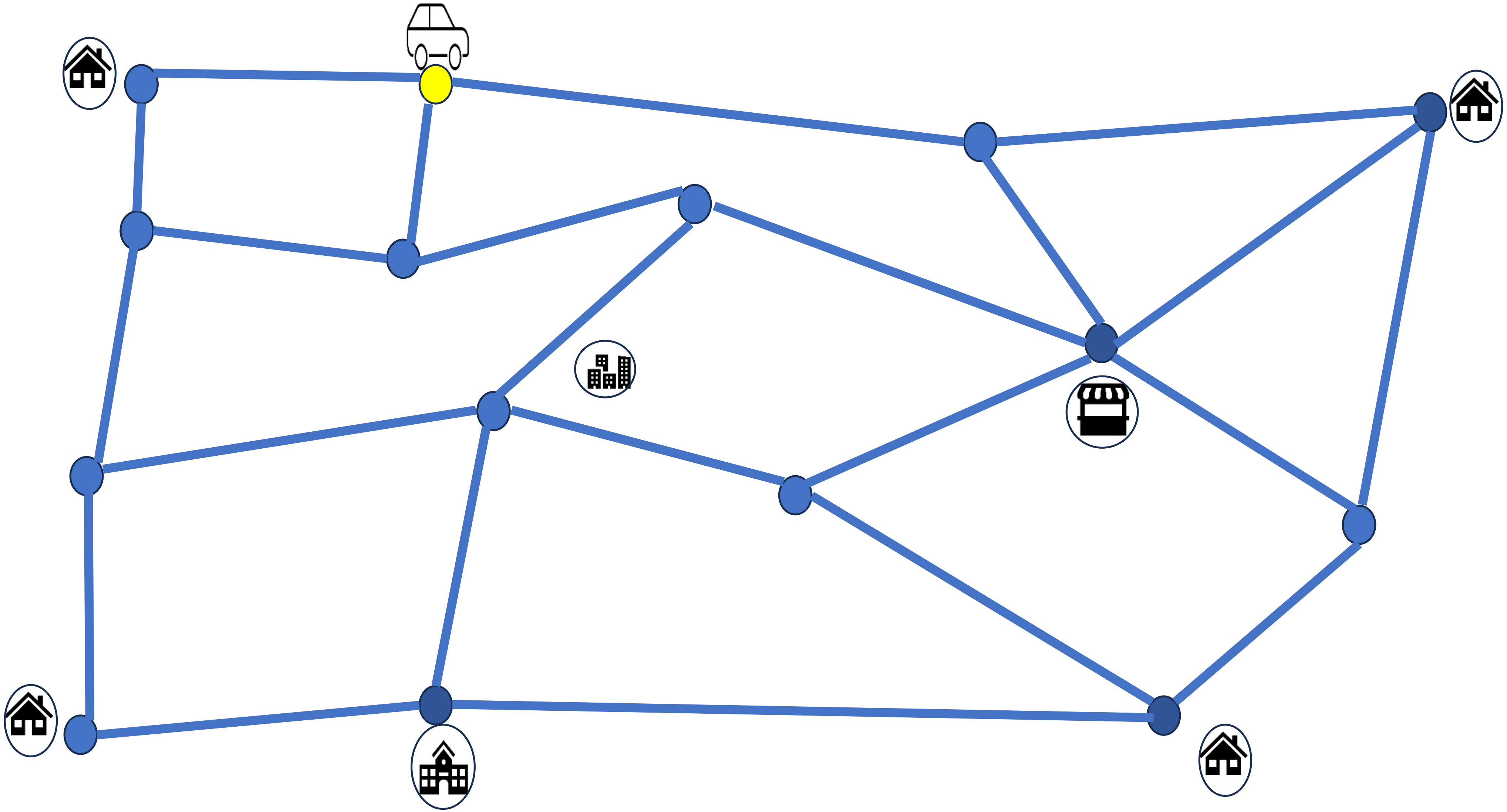
Public bus

ON-DEMAND TRANSIT

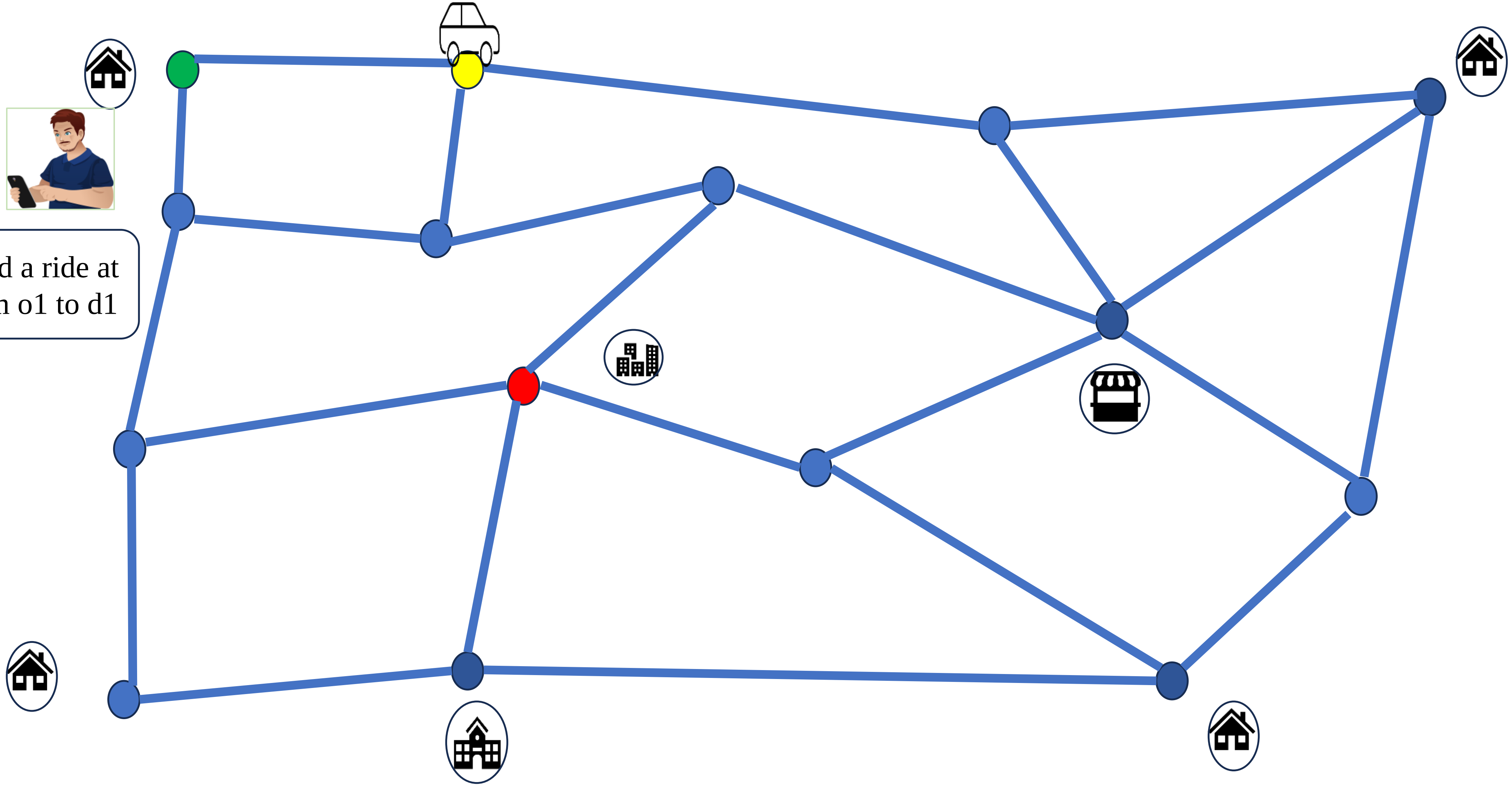
On-Demand Transit (ODT)

A shared service where vehicles update routes dynamically based on real-time traveler requests.

ON-DEMAND TRANSIT

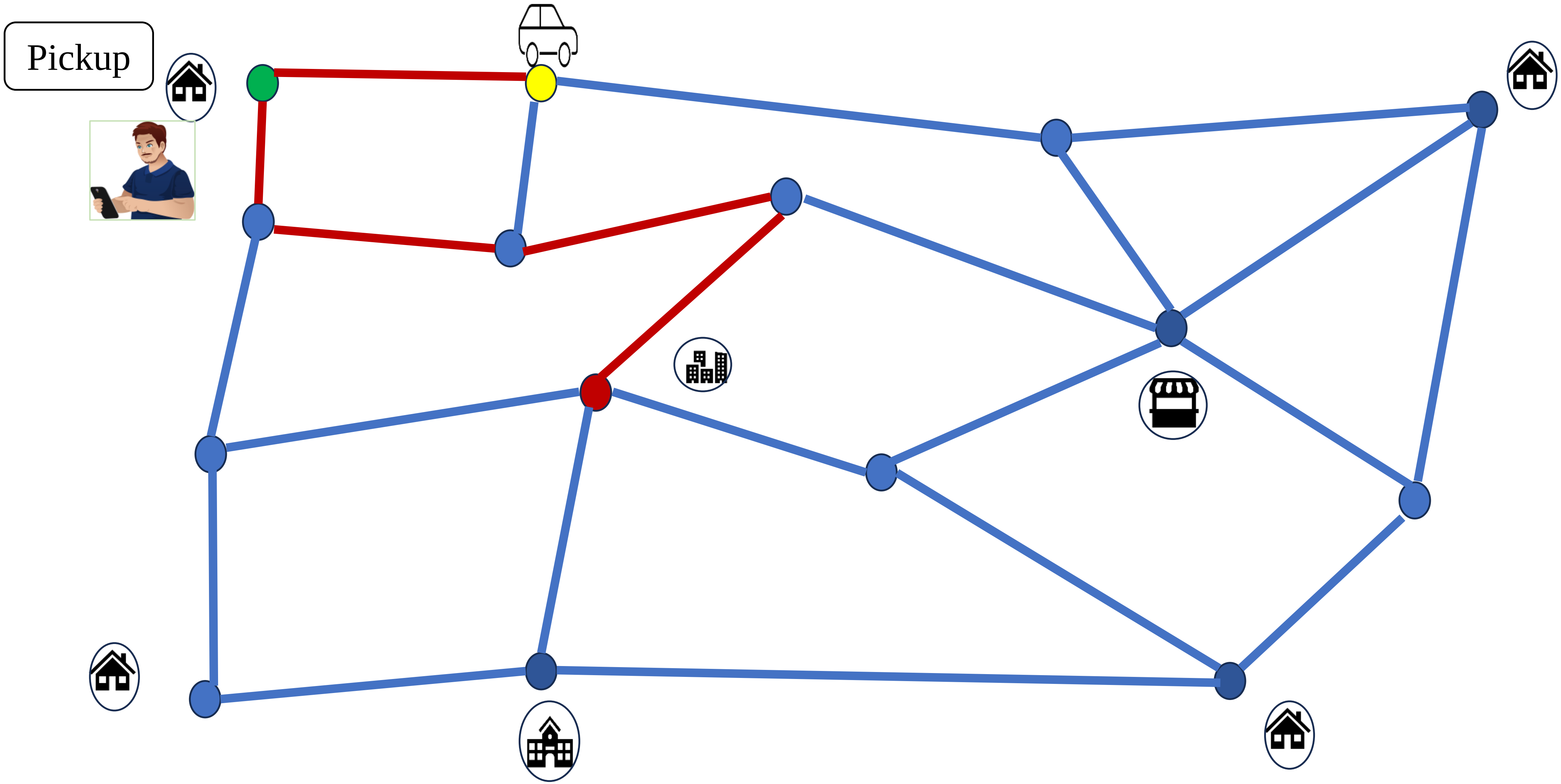


ON-DEMAND TRANSIT

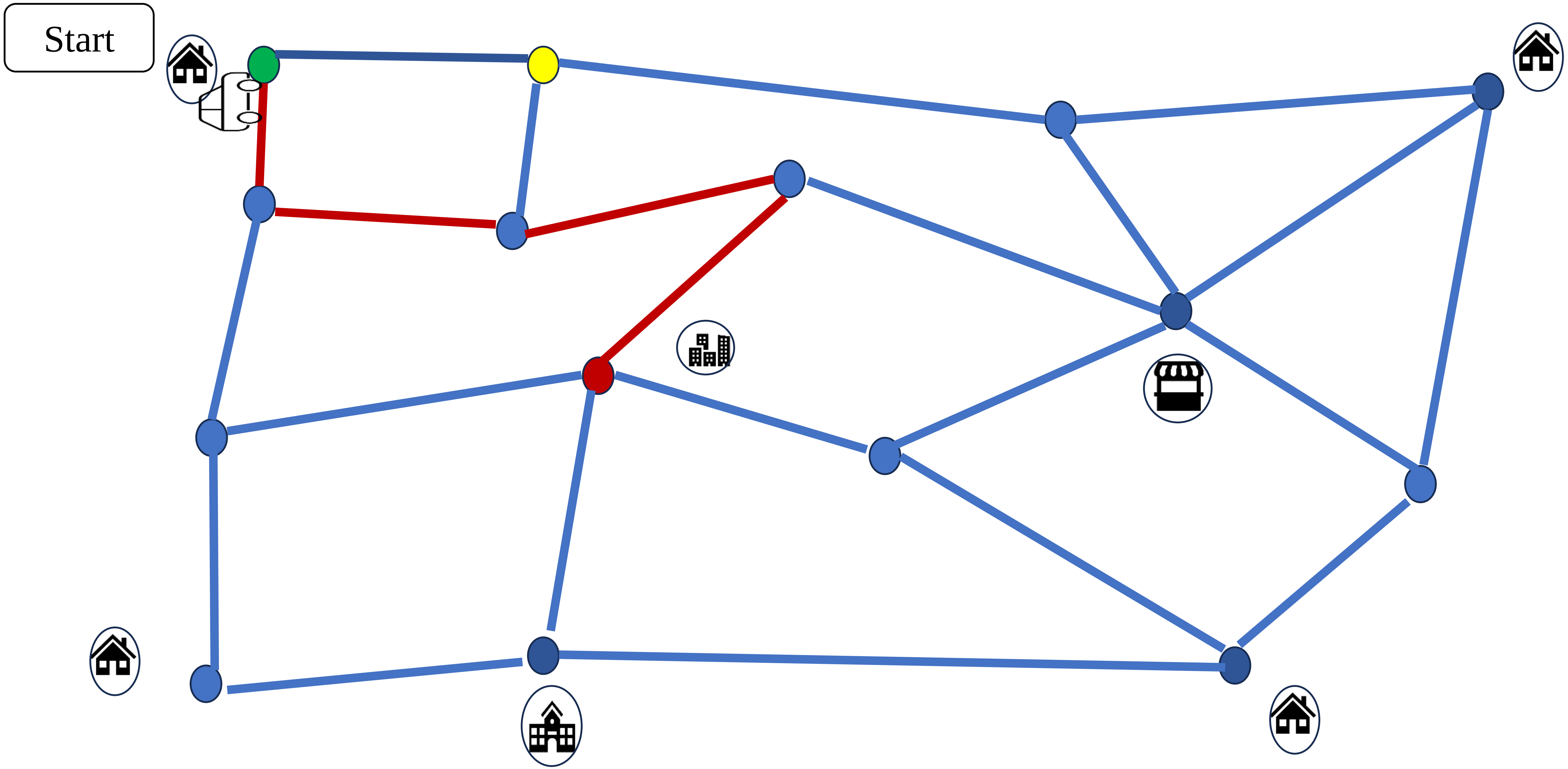


Hi! I need a ride at 8am from o1 to d1

ON-DEMAND TRANSIT



ON-DEMAND TRANSIT



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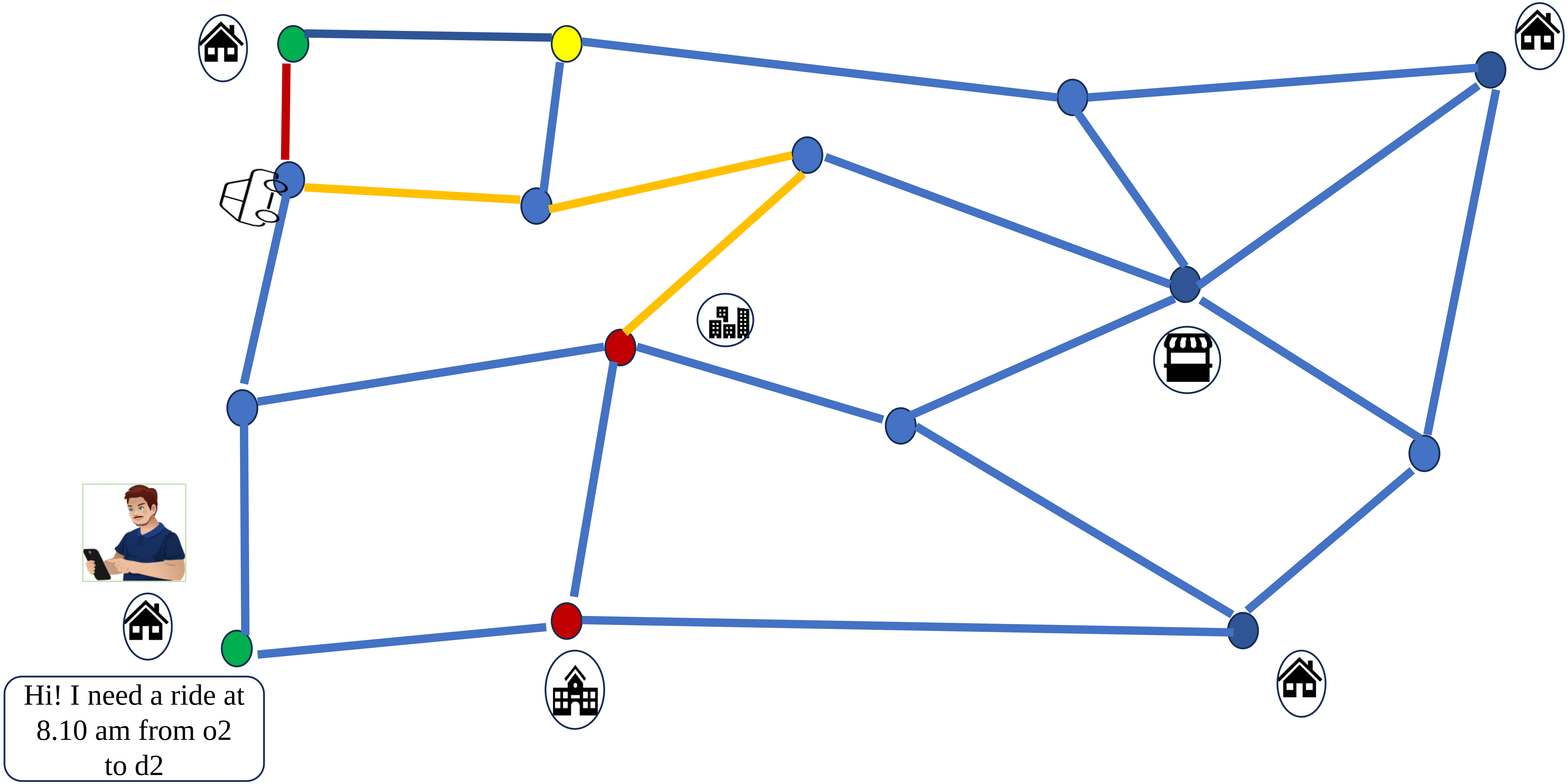
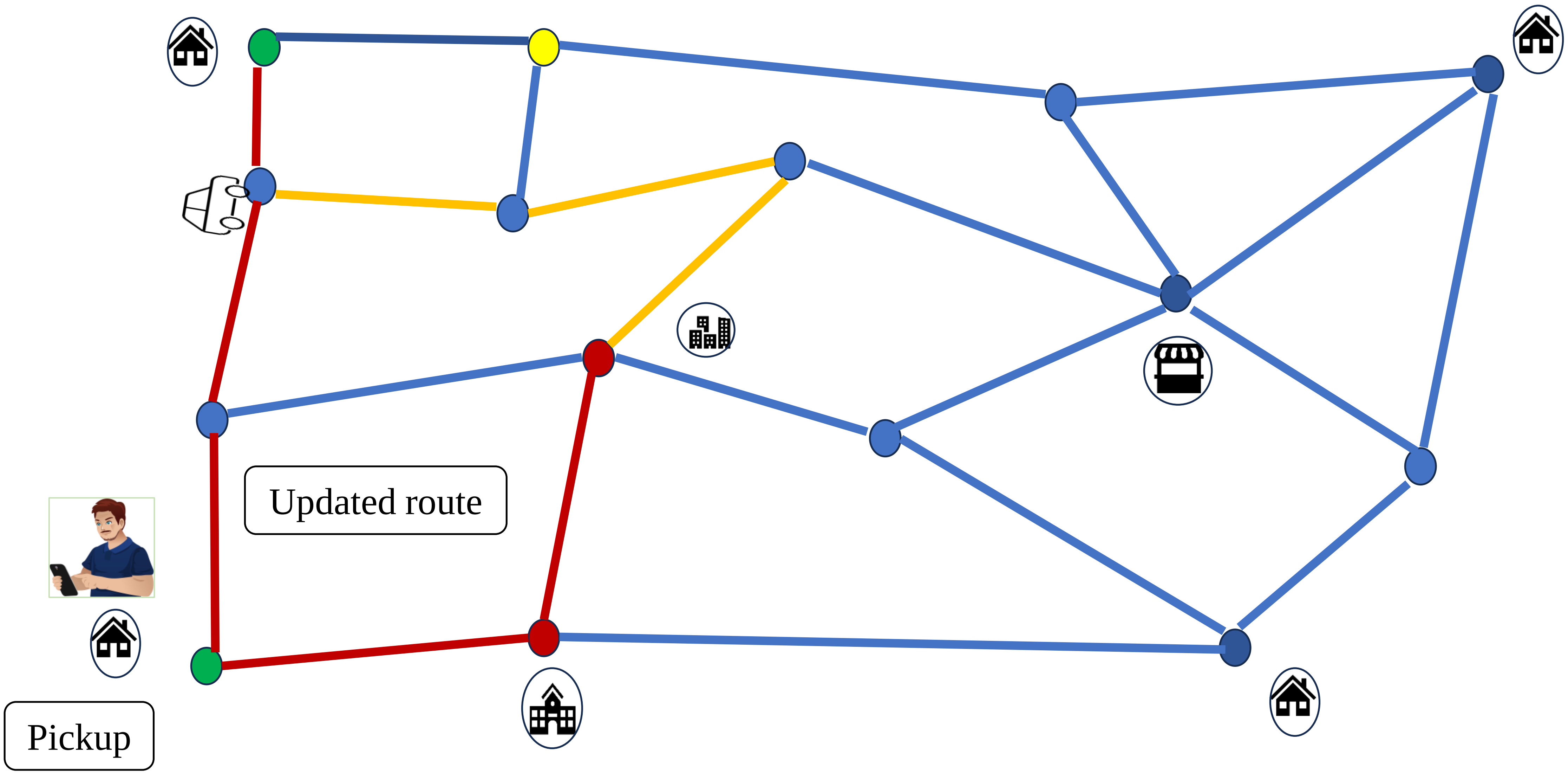
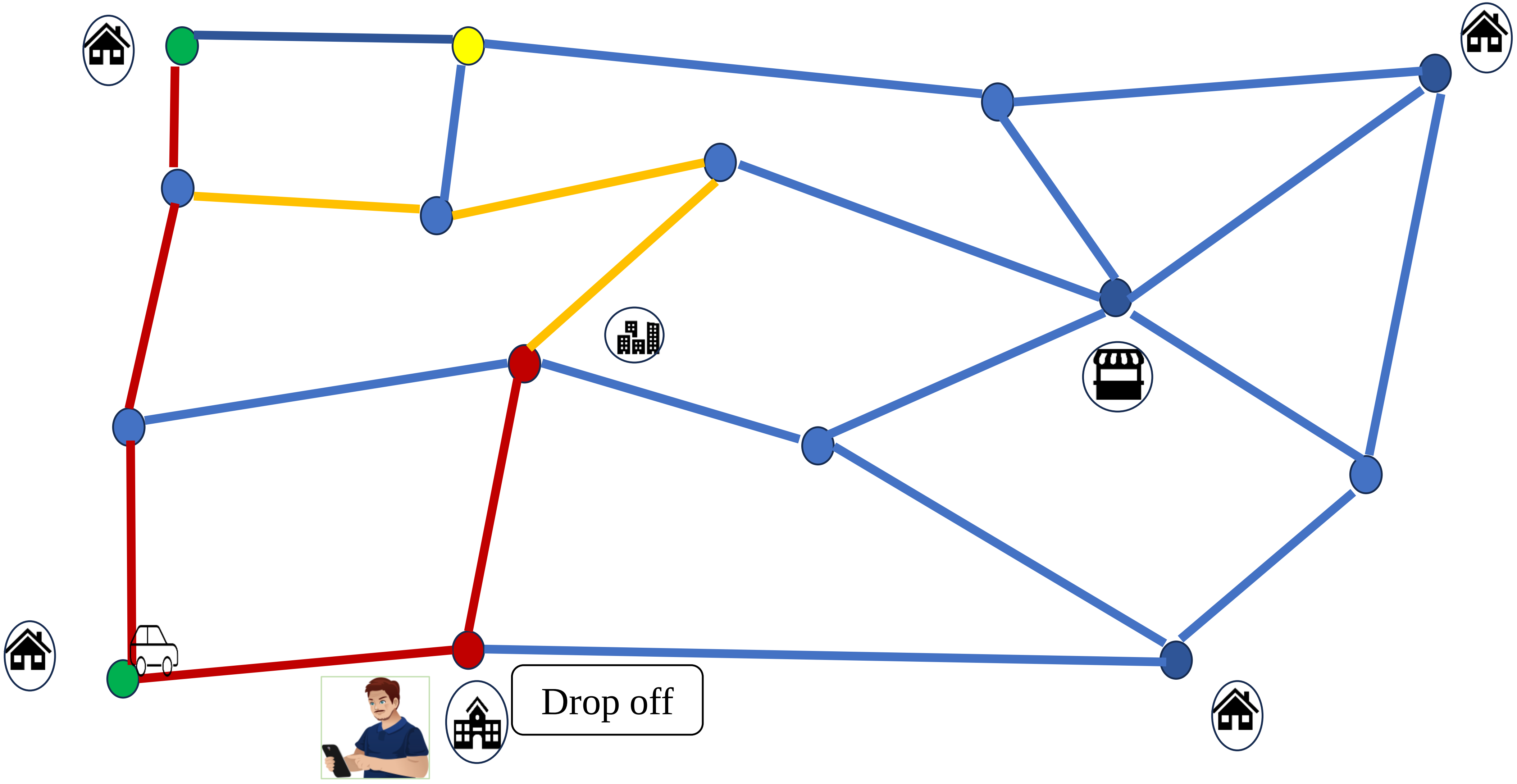


ON-DEMAND TRANSIT



ON-DEMAND TRANSIT



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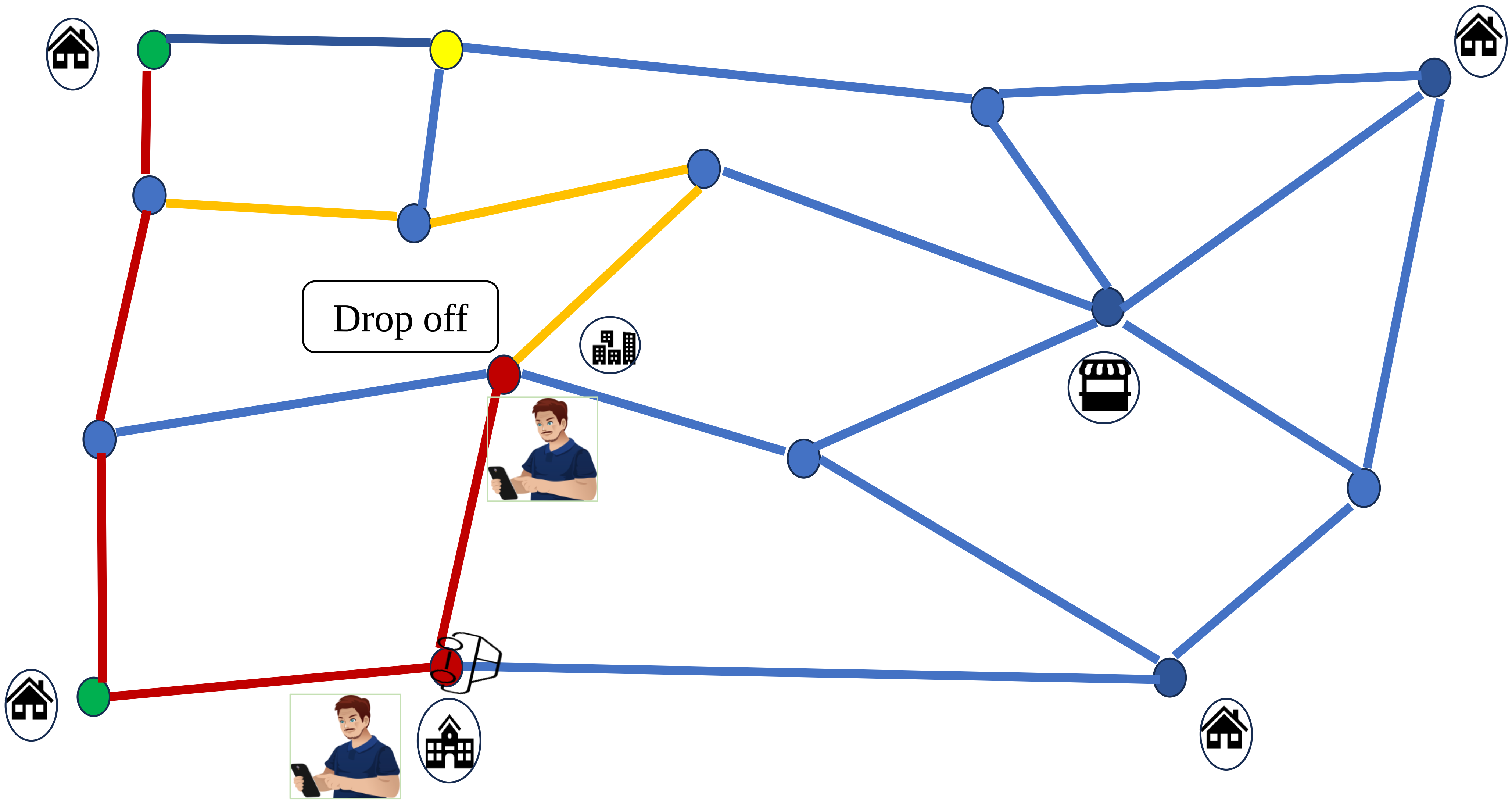
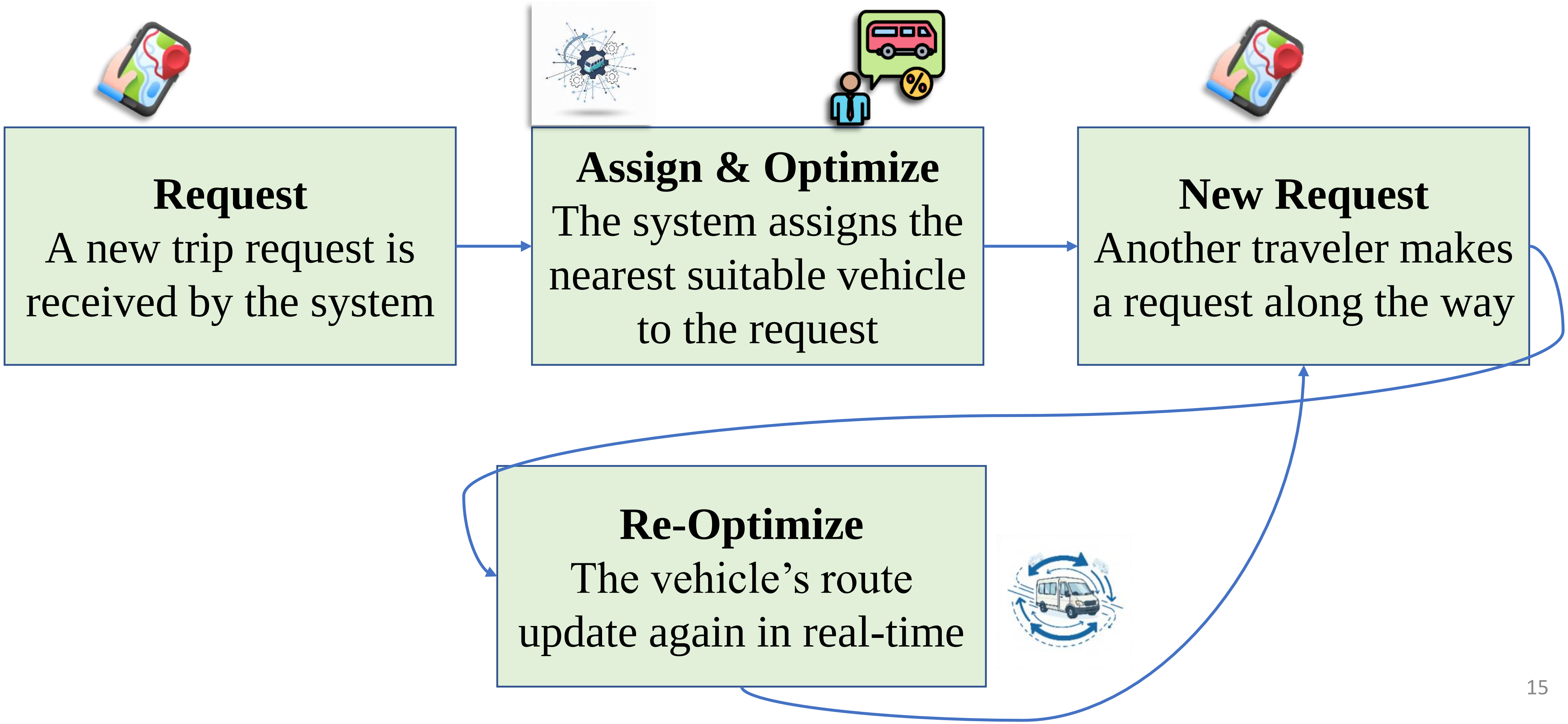


ON-DEMAND TRANSIT



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LITERATURE REVIEW

Formulation	Algorithm	Network Size	Route	Reference
VRP	Simulated annealing	Small	Fixed	Zidi *et al.*2012
VRP	Simulated annealing with GA	Small	Fixed	Huang, D. *et al.* 2020
VRP	Hybrid genetic algorithm	Small	Fixed	Masmoudi *et al.* 2017
Road network design	Branch & Bound	Small	Fixed	Huang, A. *et al.* 2020
Integer programming	Branch & Bound	Small	Fixed	Cao *et al.* 2017

LIMITATION

01
- Predefined route (Huang A. *et al*. 2020, Masmoudi *et al*. 2017)

02
- Small network (Huang D. *et al*. 2020, Masmoudi *et al*. 2017)

03
- Hypothetical data (Huang D. *et al*. 2020)

04
- Small data (Huang D. *et al*. 2020)

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OBJECTIVES AND OUTCOMES

Developing a simulator for **door-to-door service**

Simulation with real world dataset

Agent based simulator for on-demand **door-to-door service**

OBJECTIVES AND OUTCOMES

TSP and VRP based formulation

TSP with **simulated annealing** for trip assignment

Implementing **heuristic based algorithm** for trip assignment

VRPTW with **Hybrid Genetic Algorithm** for trip assignment

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SIMULATOR

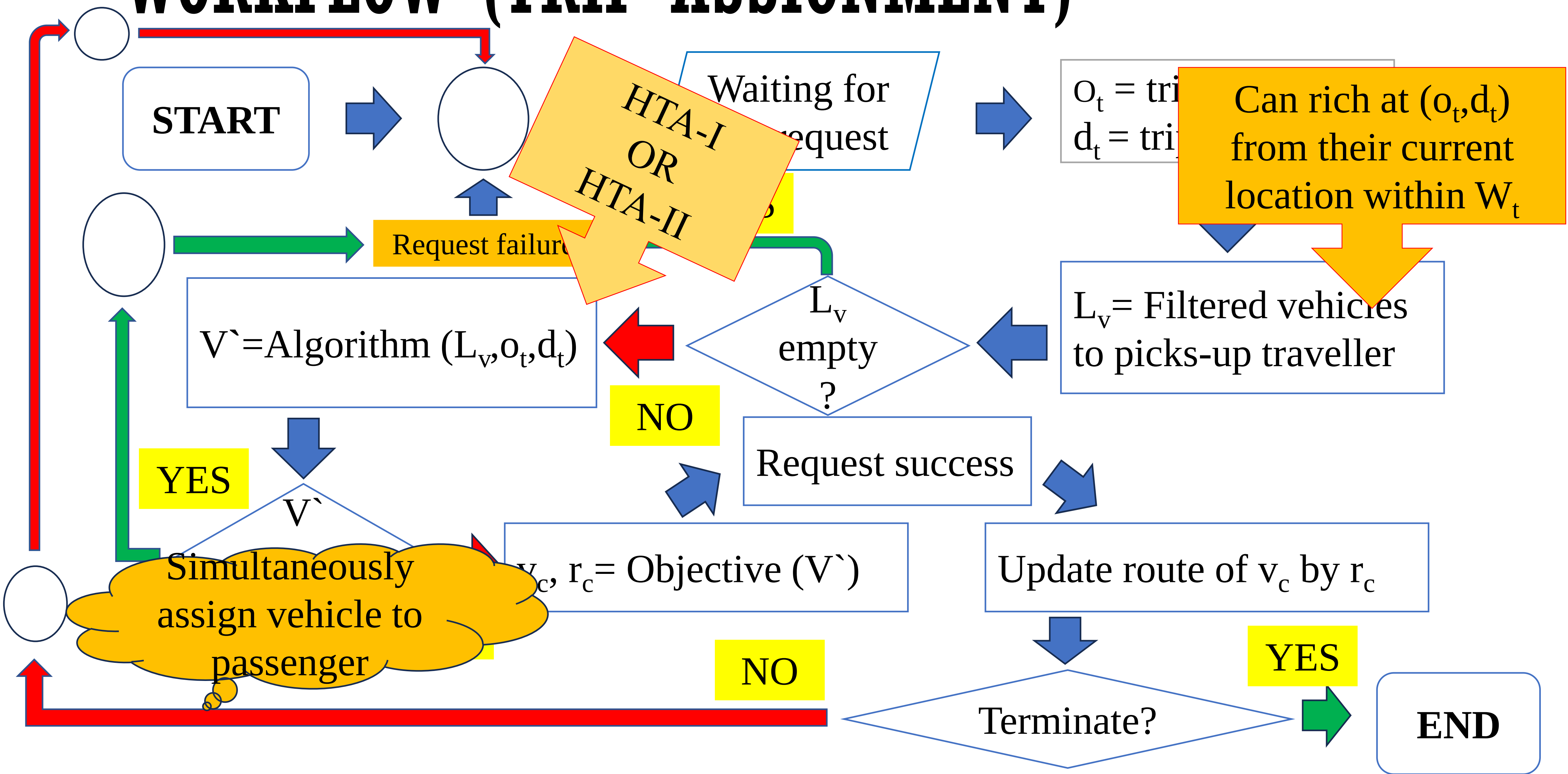
Concept of the simulator wats taken from iDAMS (**Anik et al. 2024**)

Traveler agent	Vehicle agent
At Origin	Idle
Requesting Trip	Serving Trips
Waiting for Pickup	Picking Up Passengers
In Vehicle	Dropping Off Passengers
At Destination	Route Update

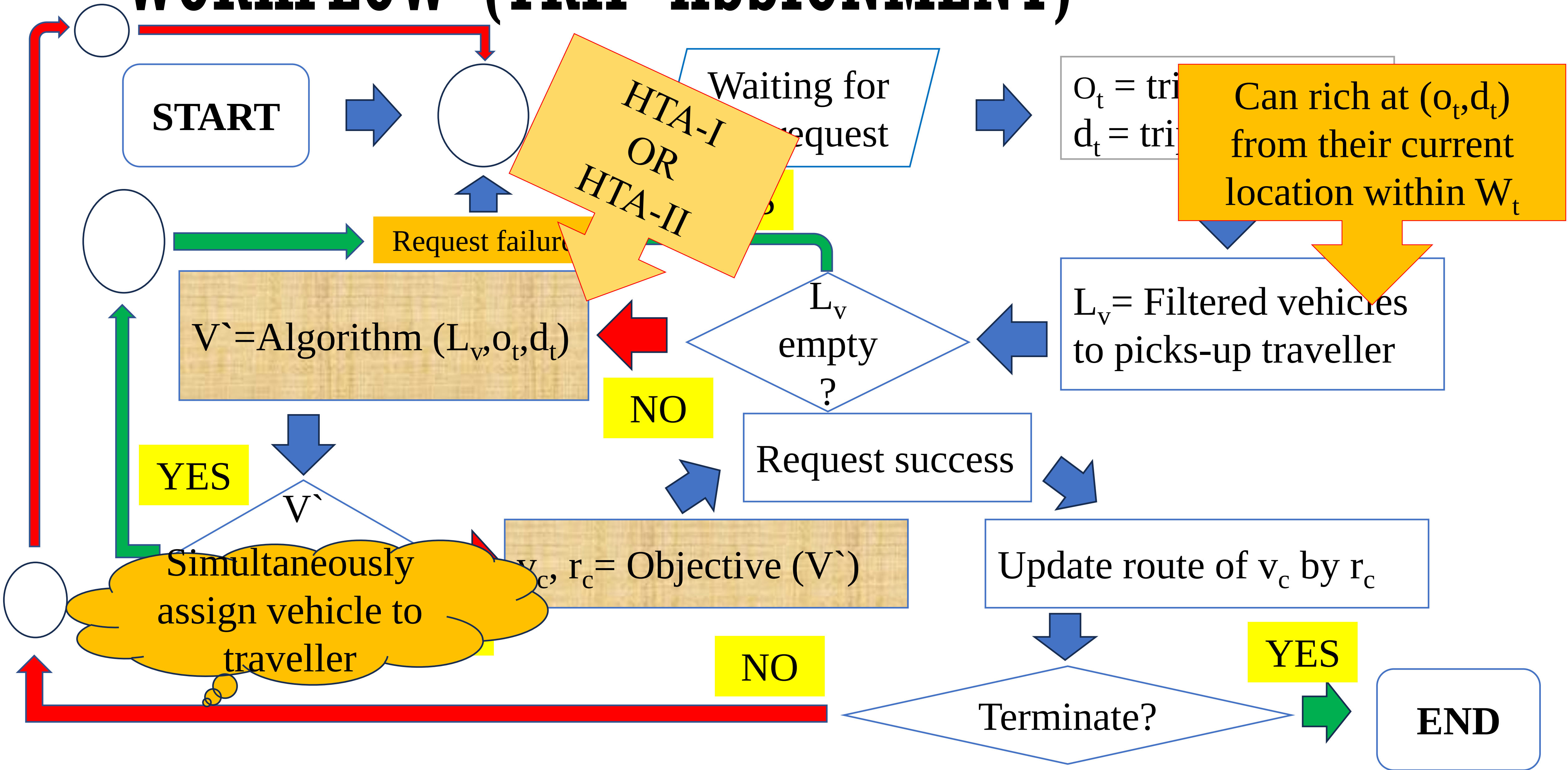
METHOD

How does the system efficiently manage large-scale requests and vehicles?

WORKFLOW (TRIP ASSIGNMENT)



WORKFLOW (TRIP ASSIGNMENT)



OBJECTIVE FUNCTIONS

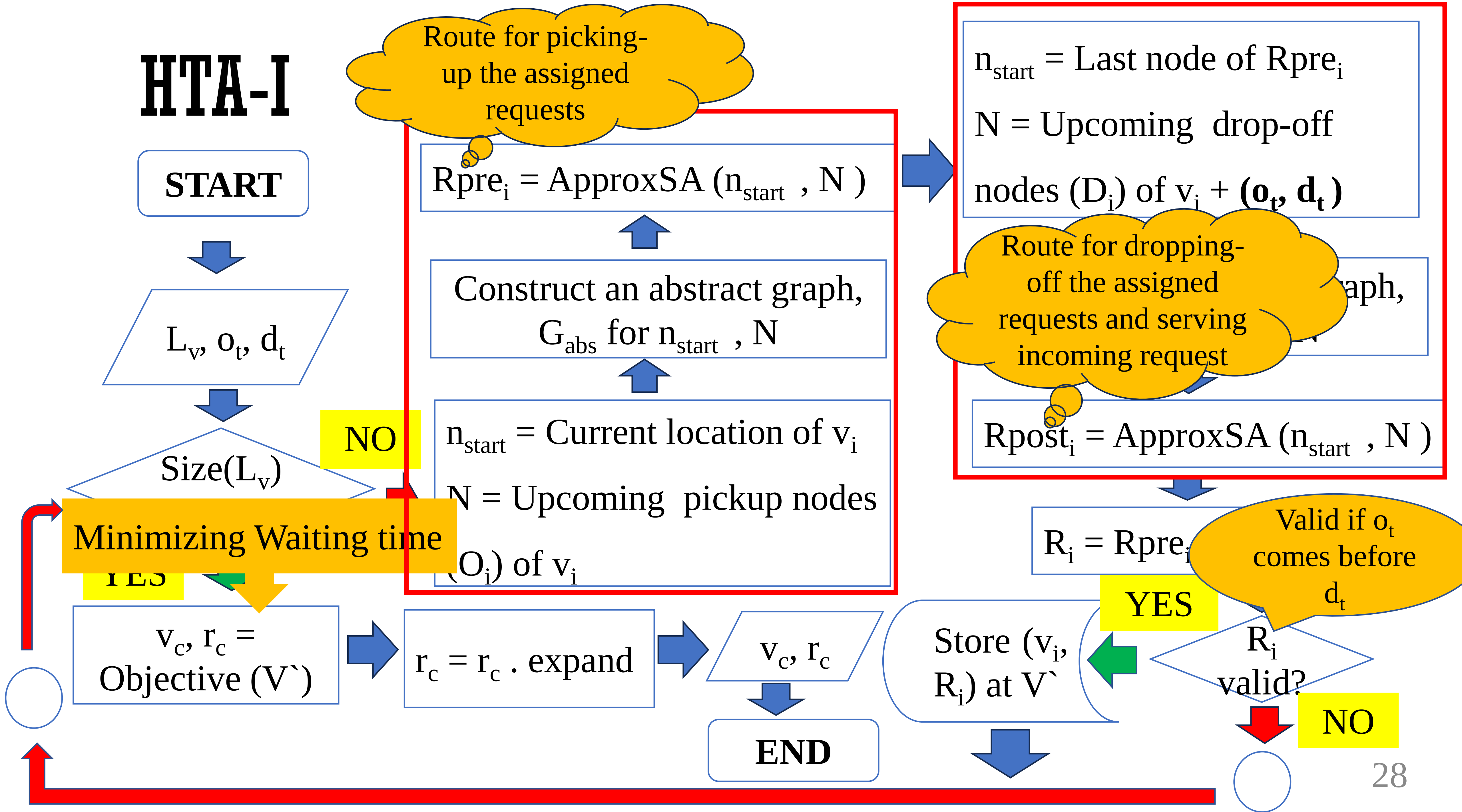
HTA-I or HTA-II result

Vehicle id	Route
v_1	R_1
v_2	R_2
--------------------	----------------------
v_i	R_i

Objective functions

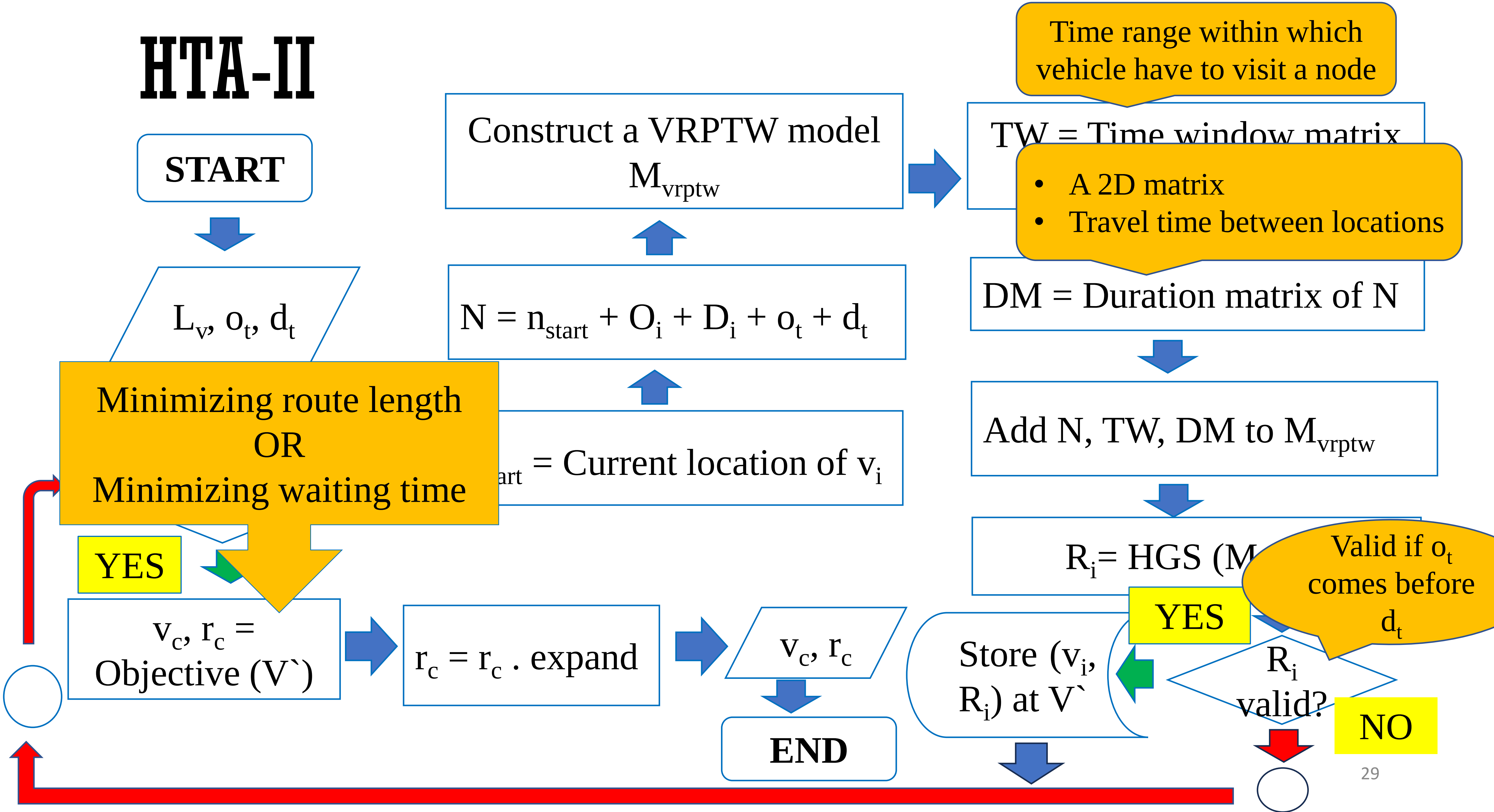
Minimizing route length	v_c = argmin (Rl_v)
Minimizing waiting time	v_c = argmin (Rwt_v)

HTA-I



Route for picking-up the assigned requests
START
L_v, o_t, d_t
Size(L_v)
NO
YES
Minimizing Waiting time
v_c, r_c = Objective (V`)
r_c = r_c . expand
v_c, r_c
END
n_{start} = Current location of v_i
N = Upcoming pickup nodes (O_i) of v_i
Construct an abstract graph, G_{abs} for n_{start} , N
$Rpre_i$ = ApproxSA (n_{start} , N)
n_{start} = Last node of $Rpre_i$
N = Upcoming drop-off nodes (D_i) of v_i + **(o_t, d_t)**
Route for dropping-off the assigned requests and serving incoming request
$Rpost_i$ = ApproxSA (n_{start} , N)
R_i = $Rpre_i$
Valid if o_t comes before d_t
R_i valid?
YES
NO
Store (v_i, R_i) at V`

HTA-II



START
L_v, o_t, d_t
Minimizing route length
OR
Minimizing waiting time
$\ldots_{start}$ = Current location of v_i
$N = n_{start} + O_i + D_i + o_t + d_t$
Construct a VRPTW model M_{vrptw}
Time range within which vehicle have to visit a node
TW = Time window matrix
A 2D matrix
Travel time between locations
DM = Duration matrix of N
Add N, TW, DM to M_{vrptw}
R_i= HGS (M
Valid if o_t comes before d_t
YES
R_i valid?
NO
Store (v_i, R_i) at V`
YES
v_c, r_c = Objective (V`)
$r_c = r_c$. expand
v_c, r_c
END

HTA-II: TIME WINDOW CALCULATION

Maximum time to visit a pickup node from vehicle current position

Node type	Time window
Vehicle current node	$(0, \infty)$
Upcoming pick-up nodes	(0, $\boxed{\text{max\_pickup\_time}}$)
Upcoming drop-off nodes	(max_pickup_time+1, ∞)
Incoming request's pick-up	(0 $\boxed{W_t}$)
Incoming request's drop-off	(W_t+1, ∞)

- **0 indicate current timestamp**
- **∞ indicate any moment**

W_t = Global waiting time for each request

HTA-II: OBJECTIVE

- **Cost function:** Time window penalty
- For each time window violation assign a time window penalty
- Minimizing this cost

VALID OR INVALID ROUTE

$n_{start} \Rightarrow o_{11} \Rightarrow o_{12} \Rightarrow o_{13} \Rightarrow d_{11} \Rightarrow d_{13} \Rightarrow o_t \Rightarrow d_{13} \Rightarrow d_t$

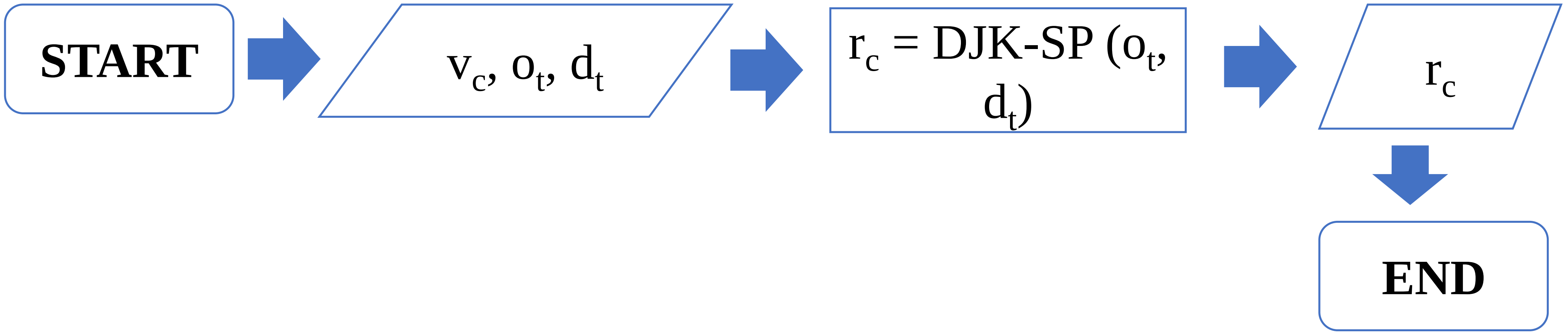
Figure: A valid route example

A drop-off node is visited prior to its corresponding pickup node

$n_{start} \Rightarrow o_{11} \Rightarrow o_{12} \Rightarrow o_{13} \Rightarrow d_{11} \Rightarrow d_{13} \Rightarrow d_t \Rightarrow d_{13} \Rightarrow o_t$

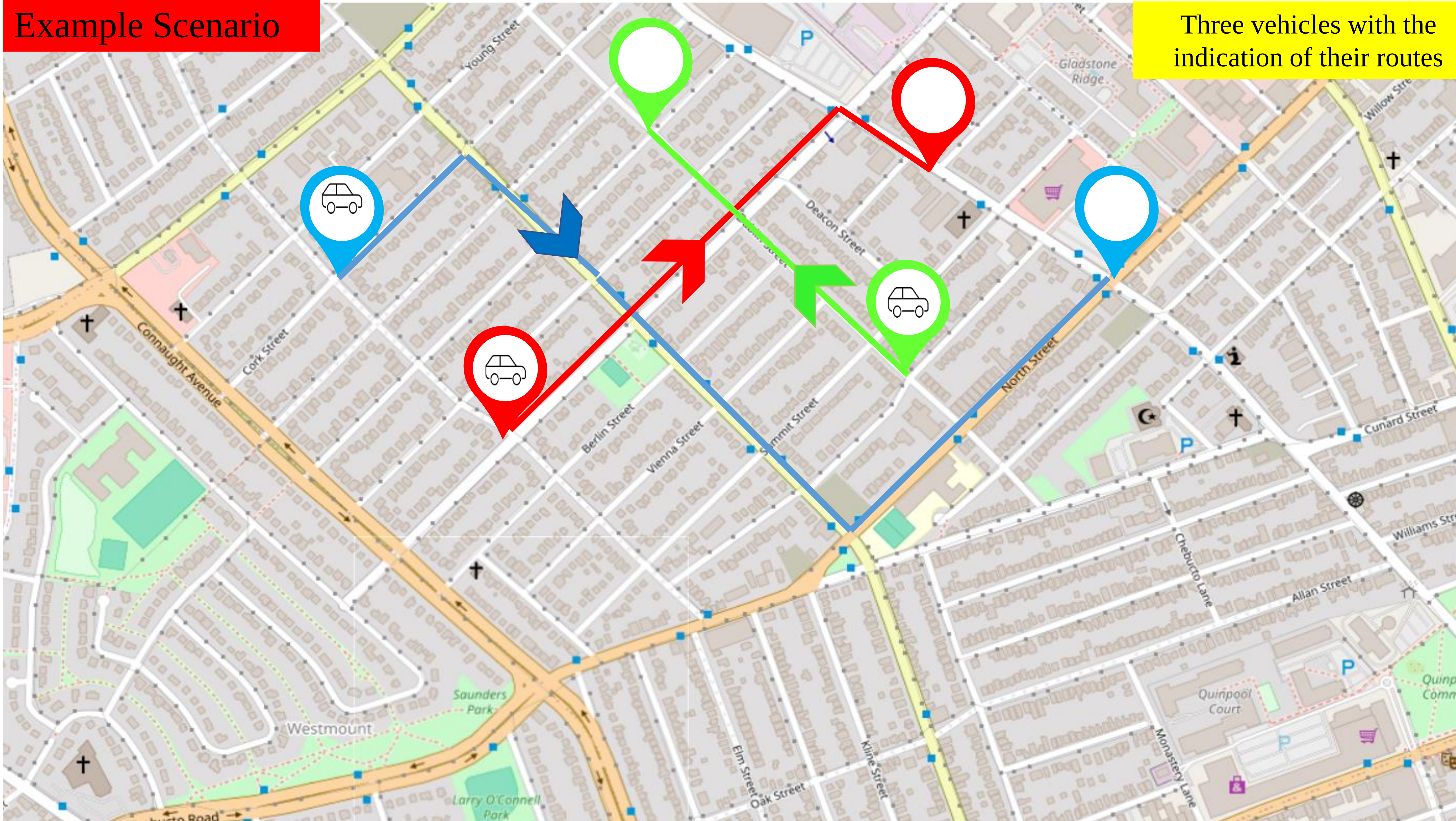
Figure: An invalid route example

BASELINE: PRIVATE CAR



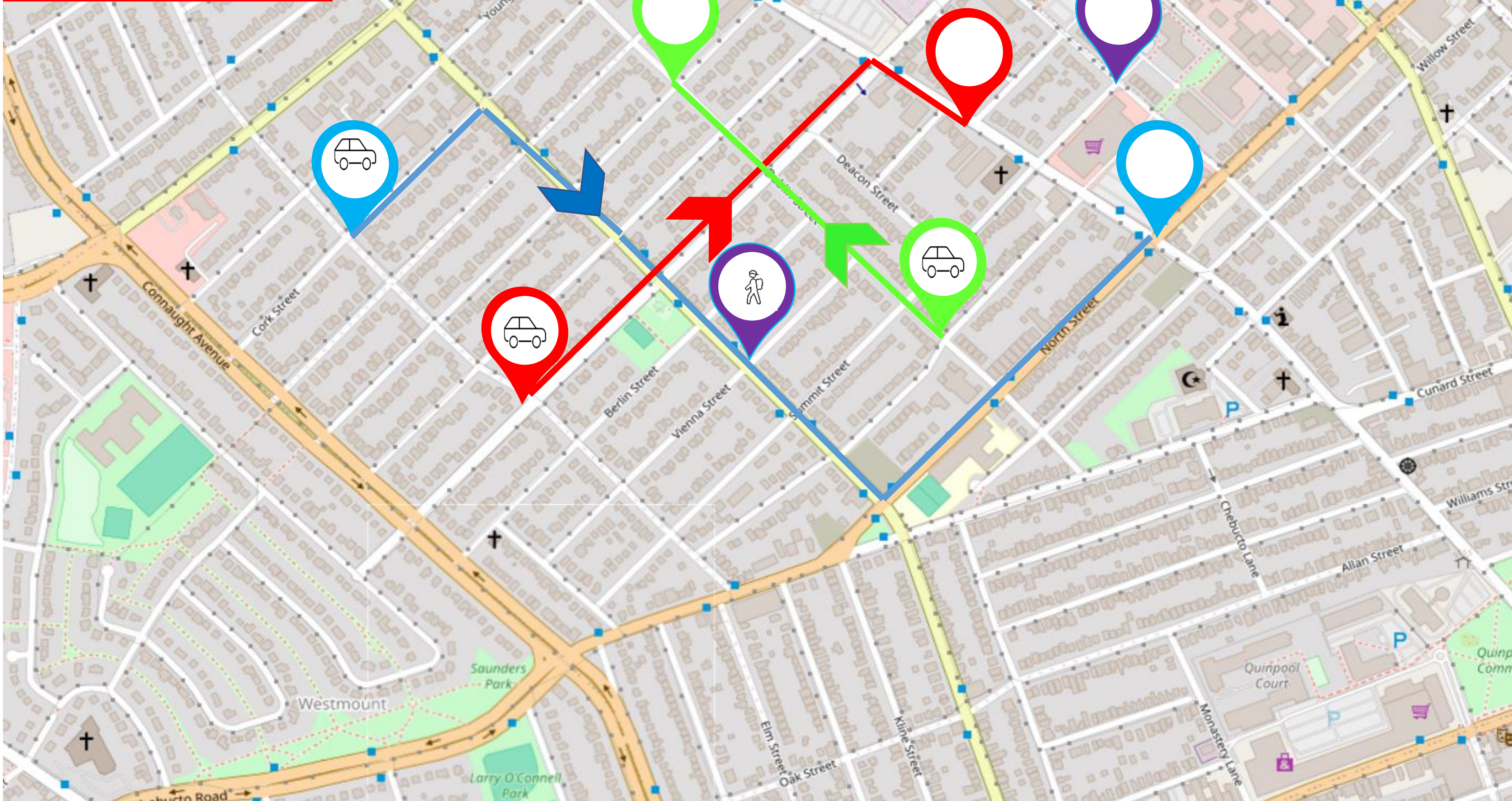
Example Scenario

Three vehicles with the indication of their routes



Example Scenario

Request come from a traveler

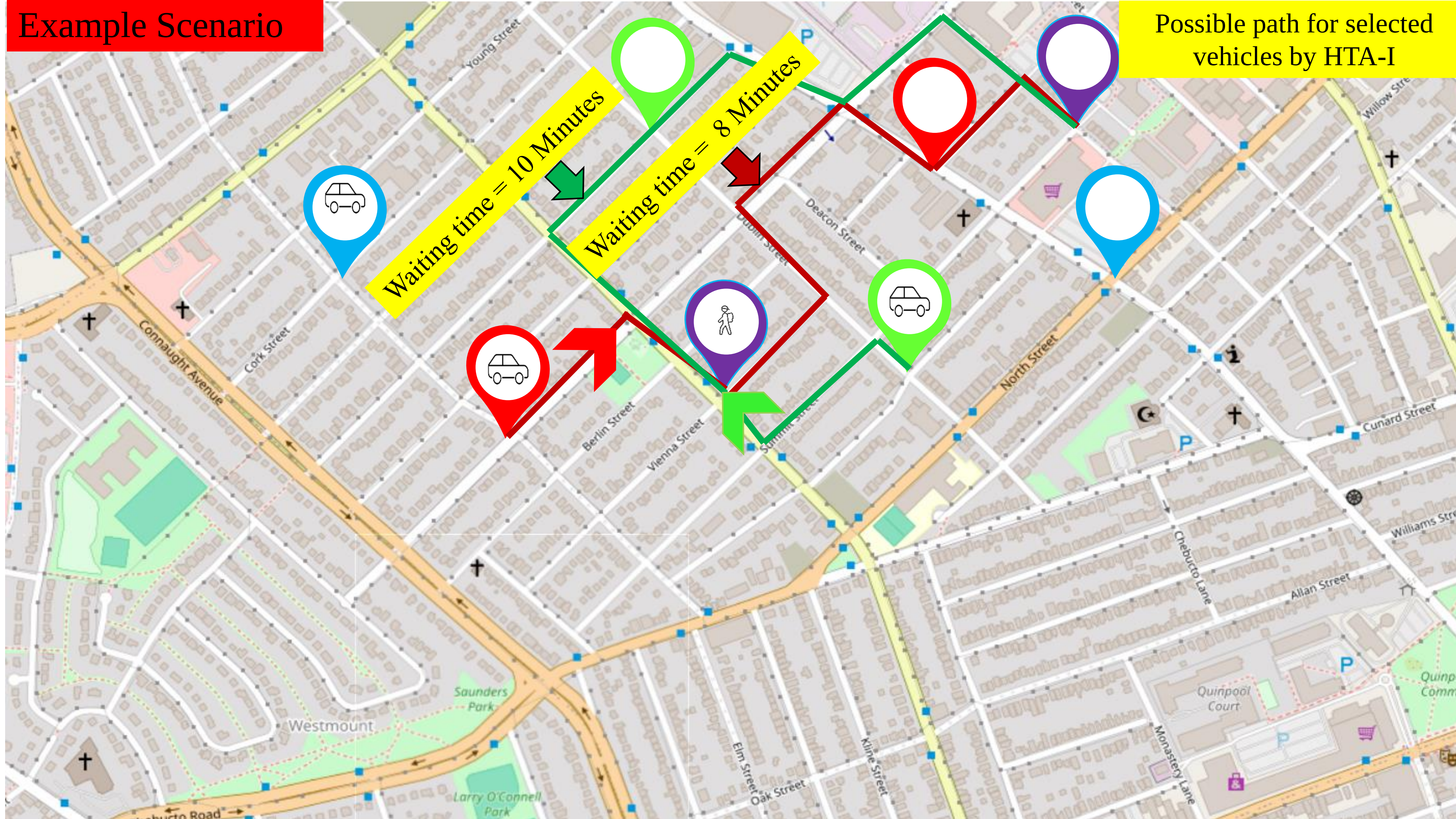


Example Scenario

Selecting nearest two vehicles

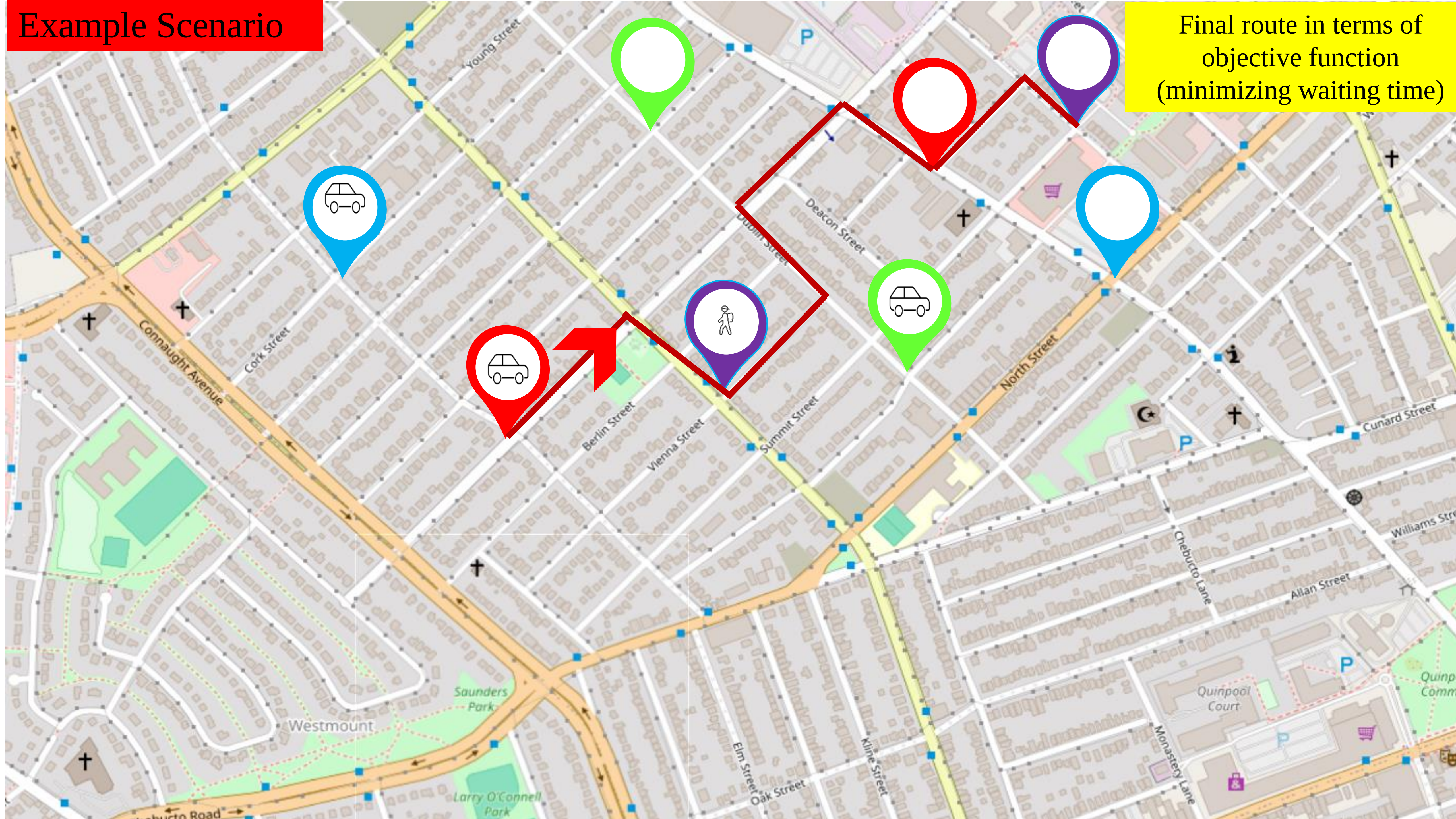
Example Scenario

Possible path for selected vehicles by HTA-I



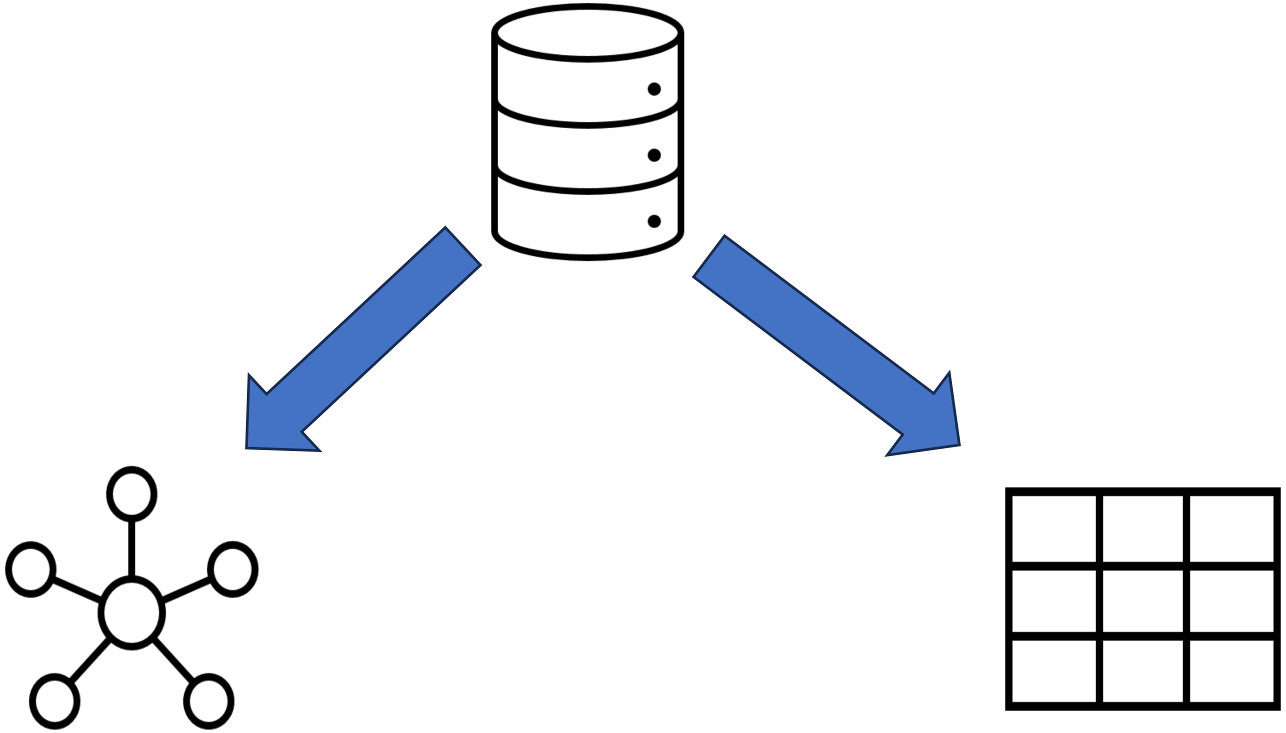
Example Scenario

Final route in terms of objective function (minimizing waiting time)



OUTLINE

DATASET



Road
network

Travel
activity

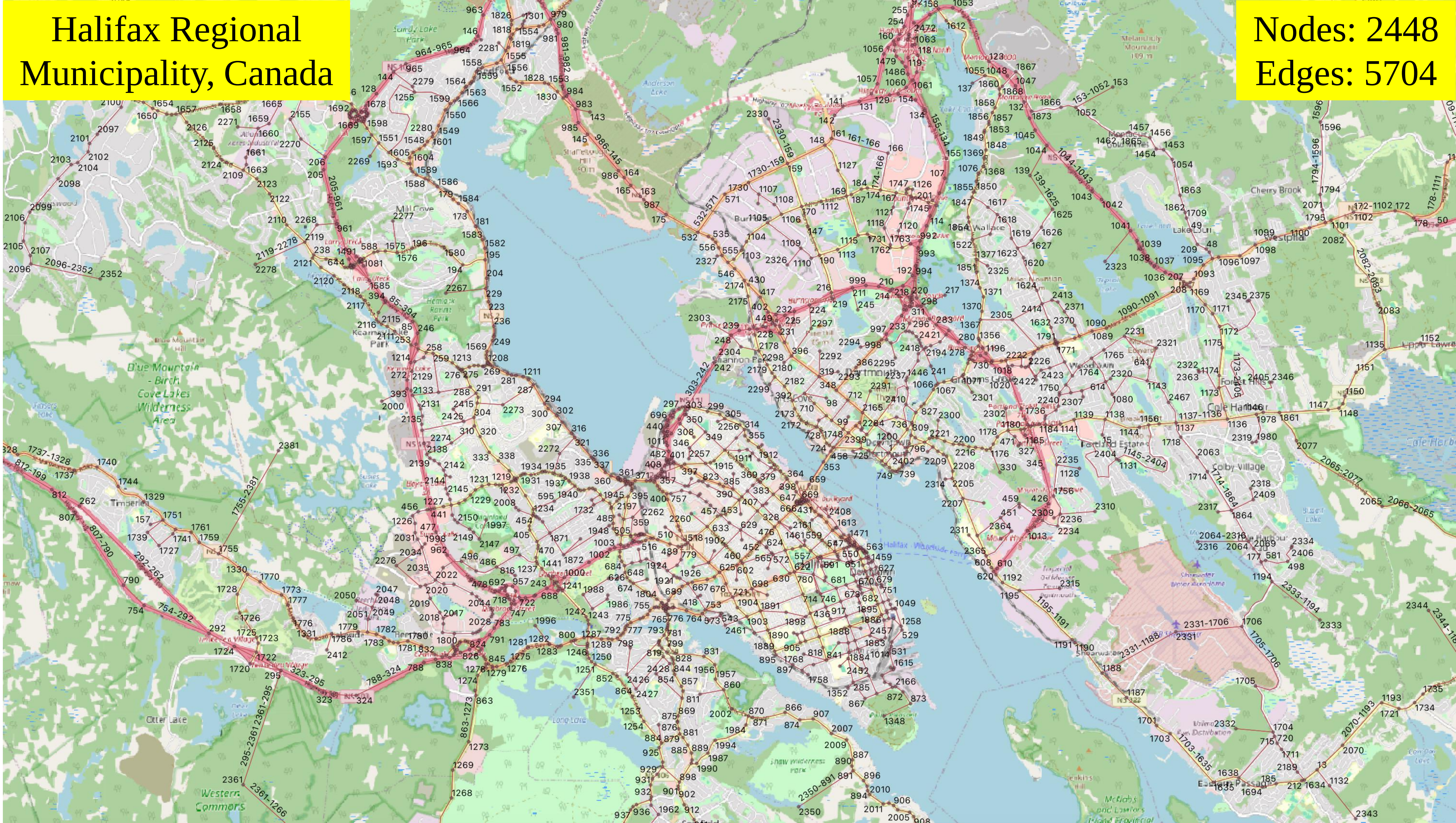
DATASET (TRIP REQUESTS)

- Nova Scotia Travel Activity (NovaTRAC) surveys
- Conducted from 2015-2018
- 24-hour trip data of a typical weekday
- Number of requests : 10005

Trip ID	Indiv. ID	Origin node	Destination node	Departure time	Arrival time
851	531	76	1679	11:27AM	11:34AM

Halifax Regional Municipality, Canada

Nodes: 2448
Edges: 5704



HALIFAX CITY SUBNETWORK

PERFORMANCE METRICS

- **Completed-trips:** % of trip completed by vehicles
- **Average individual wait:** Average waiting time of requested traveler
- **Average individual delay:** Average delay of traveler to reach at destination

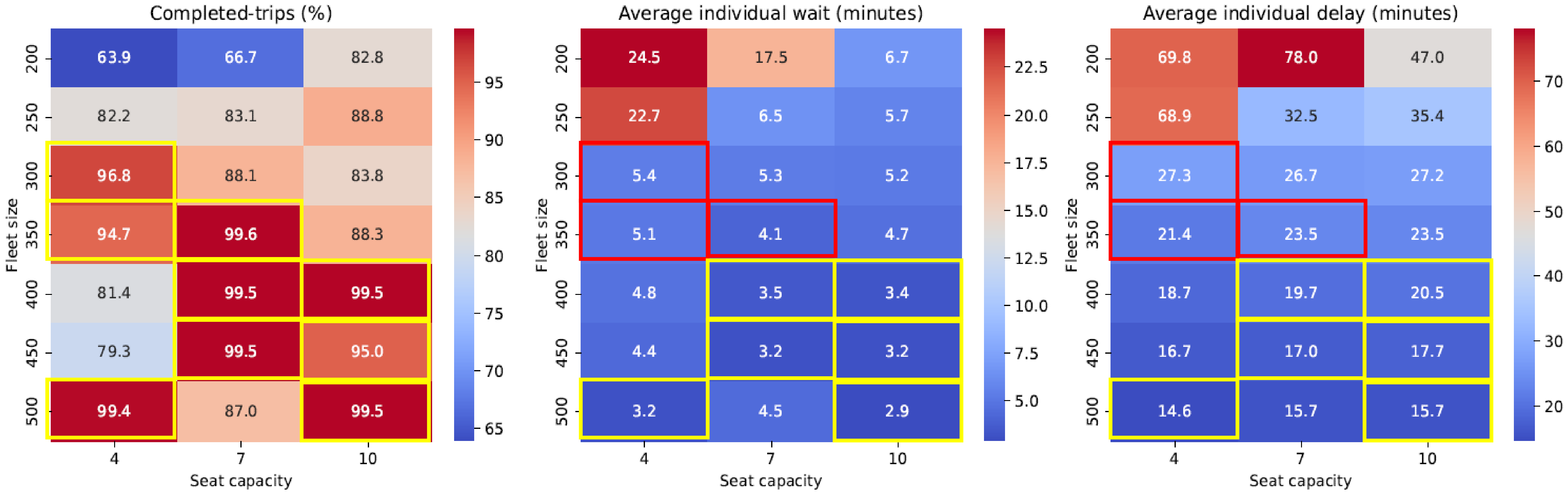
PERFORMANCE METRICS

- **VKT reduction:** % of vehicle kilometer travelled reduced than baseline model

- **Early reached:** % of travelers reached before expected arrival time

RESULT: EXPERIMENTAL SETUP

- **Fleet size:** 200, 250, 300, 350, 400, 450, 500
- **Seat capacity:** 4, 7, 10
- Run for different combinations of fleet size and seat capacity
- **Vehicle distribution:** Uniform distribution
- Assume that a central authority can meet a city's travel demand



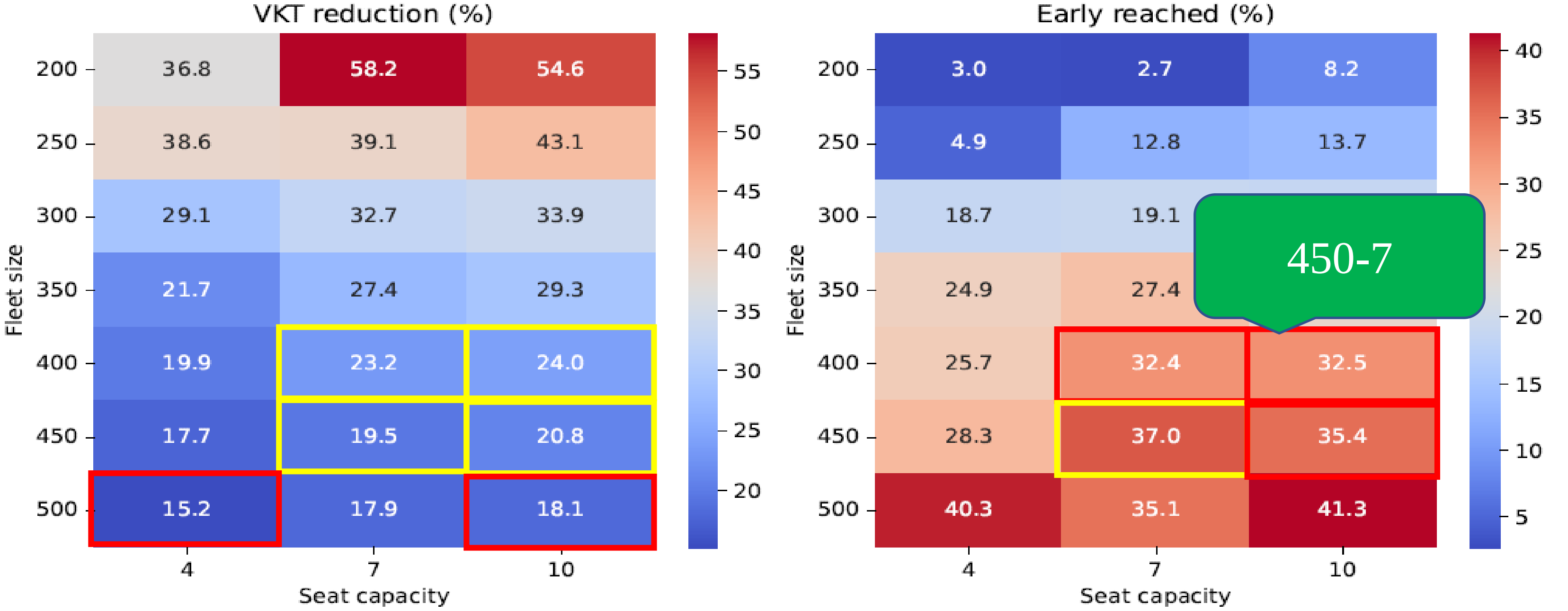
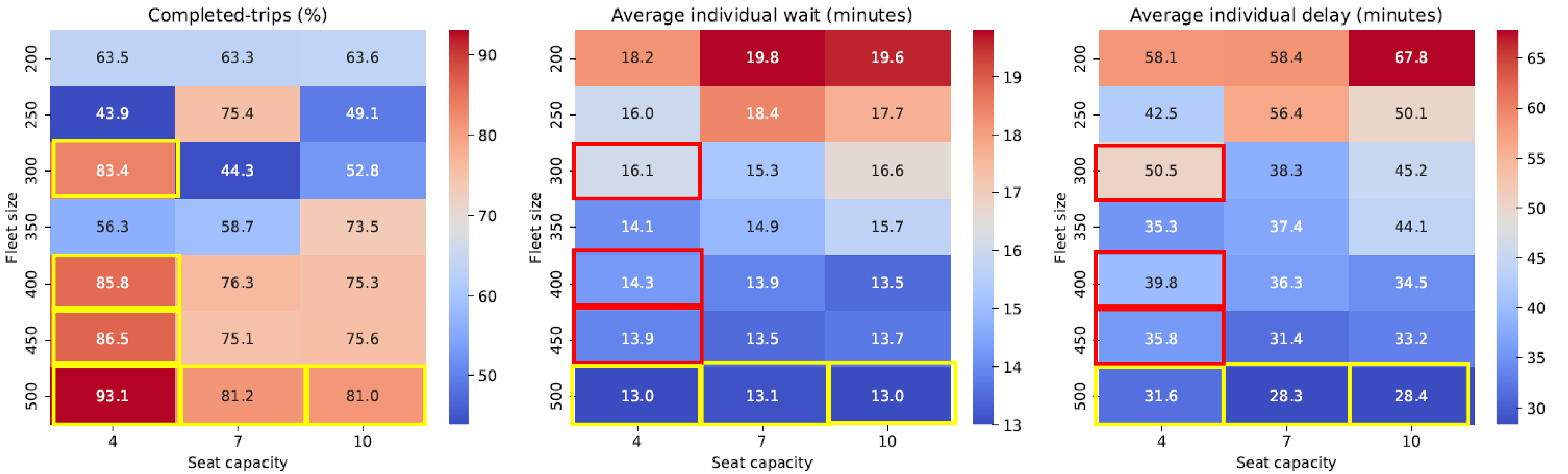


Figure: Performance analysis of HTA-I



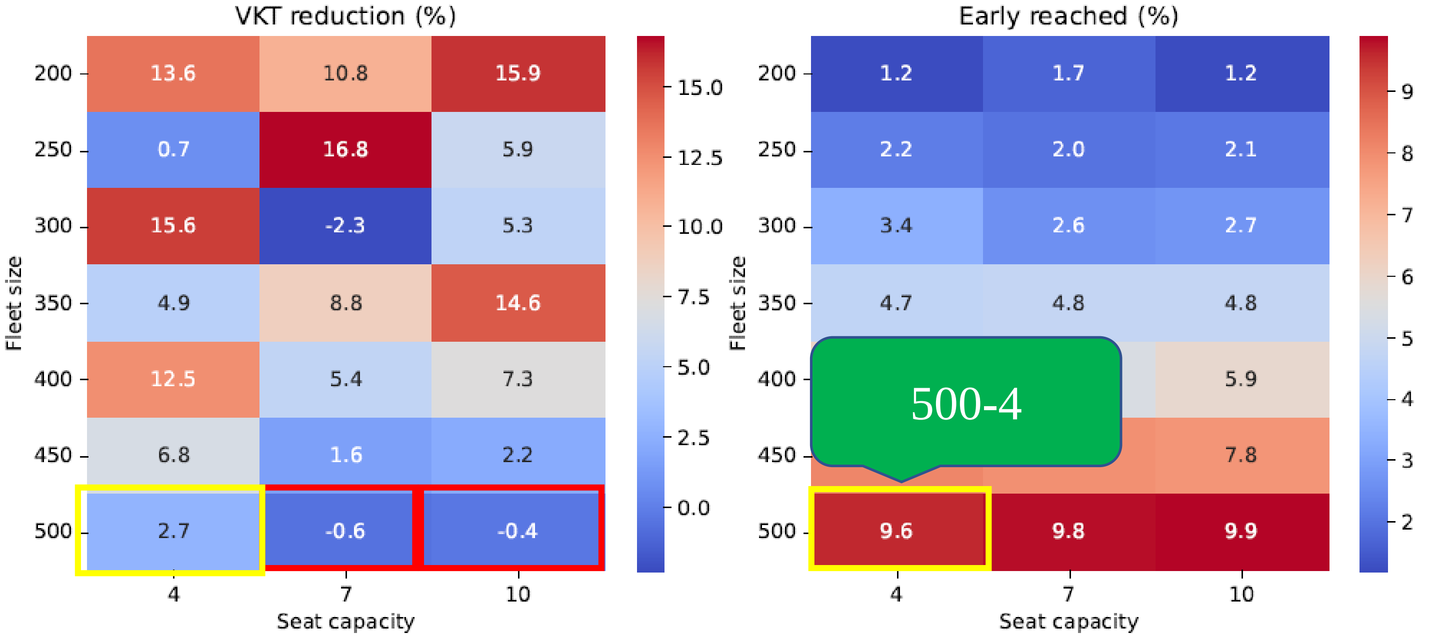
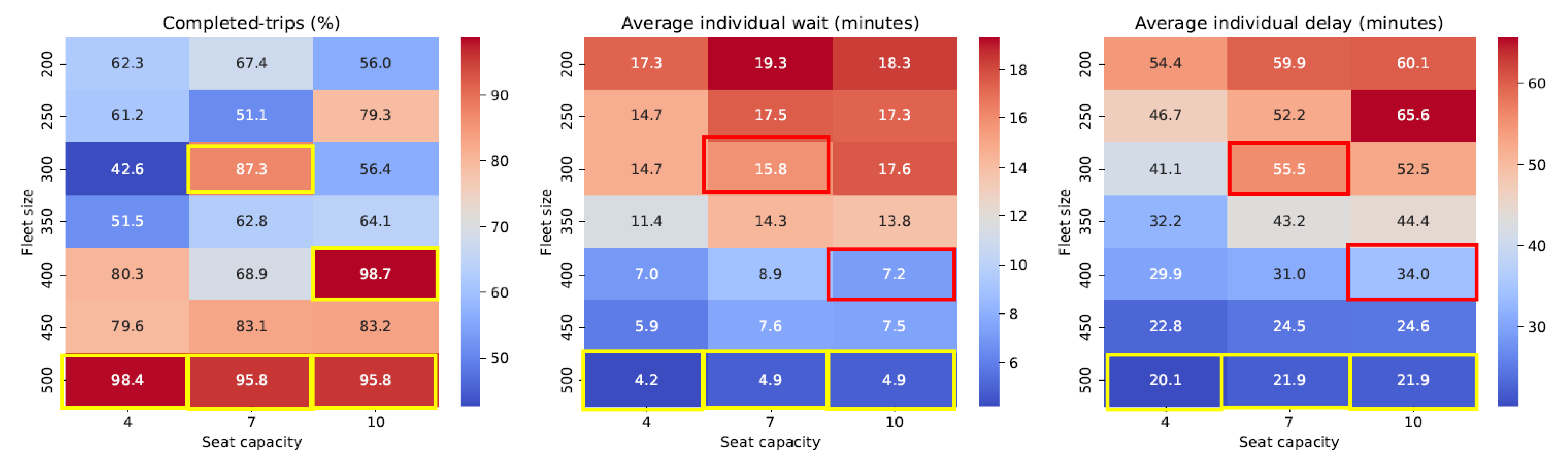


Figure: Performance analysis of HTA-II (Minimizing route length)



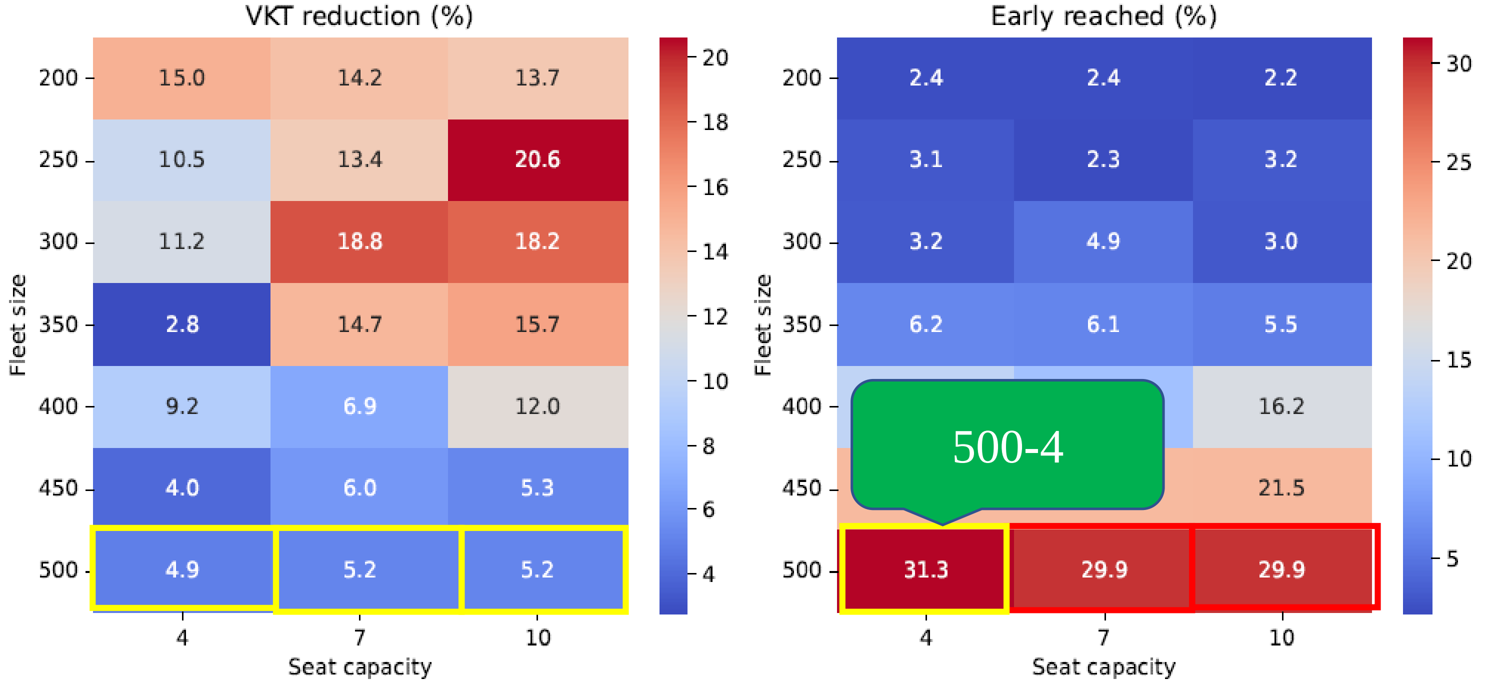
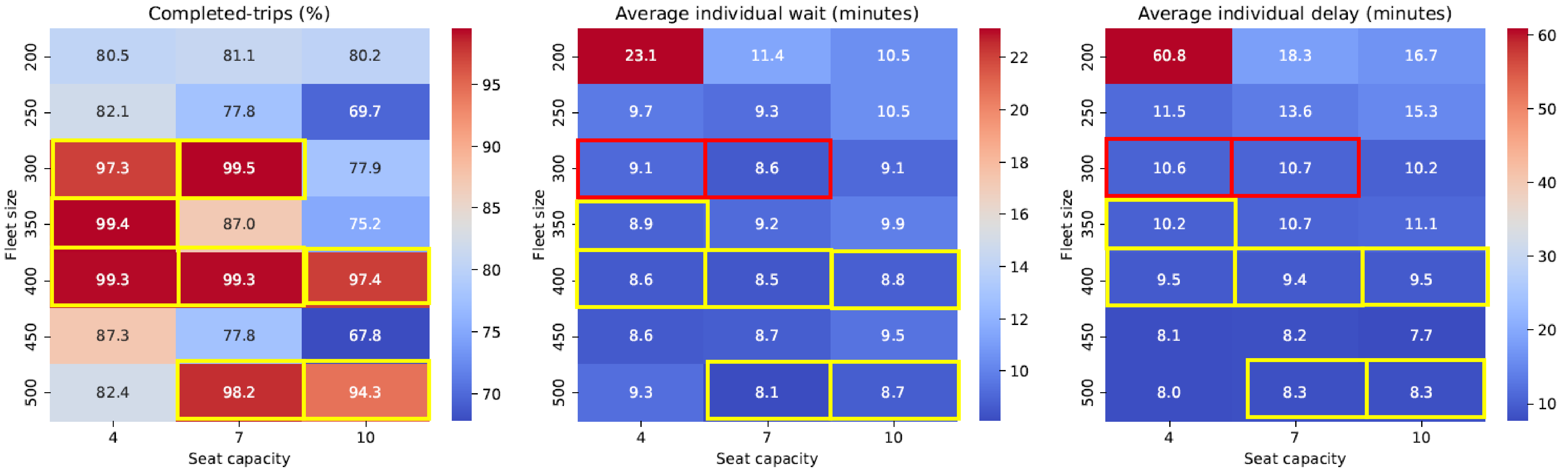


Figure: Performance analysis of HTA-II (Minimizing waiting time)



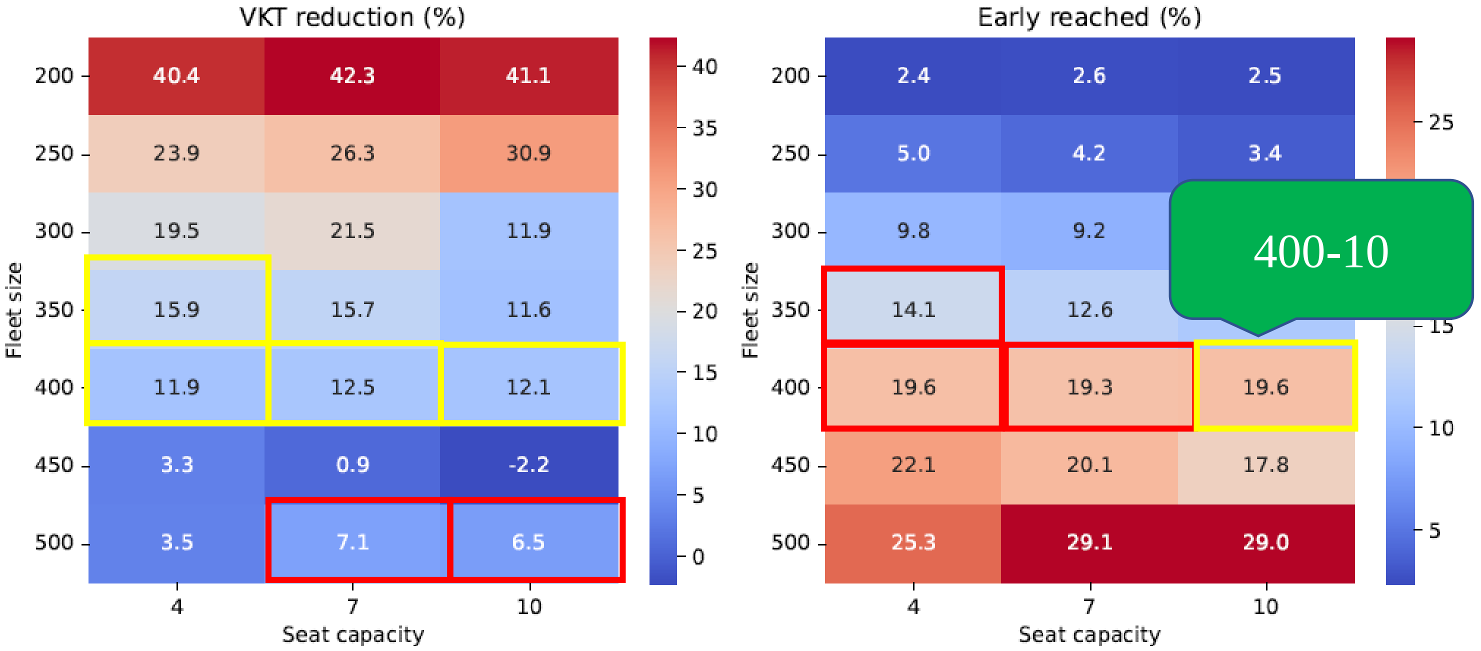


Figure: Performance analysis of iDAMS (**Anik et al. 2024**)

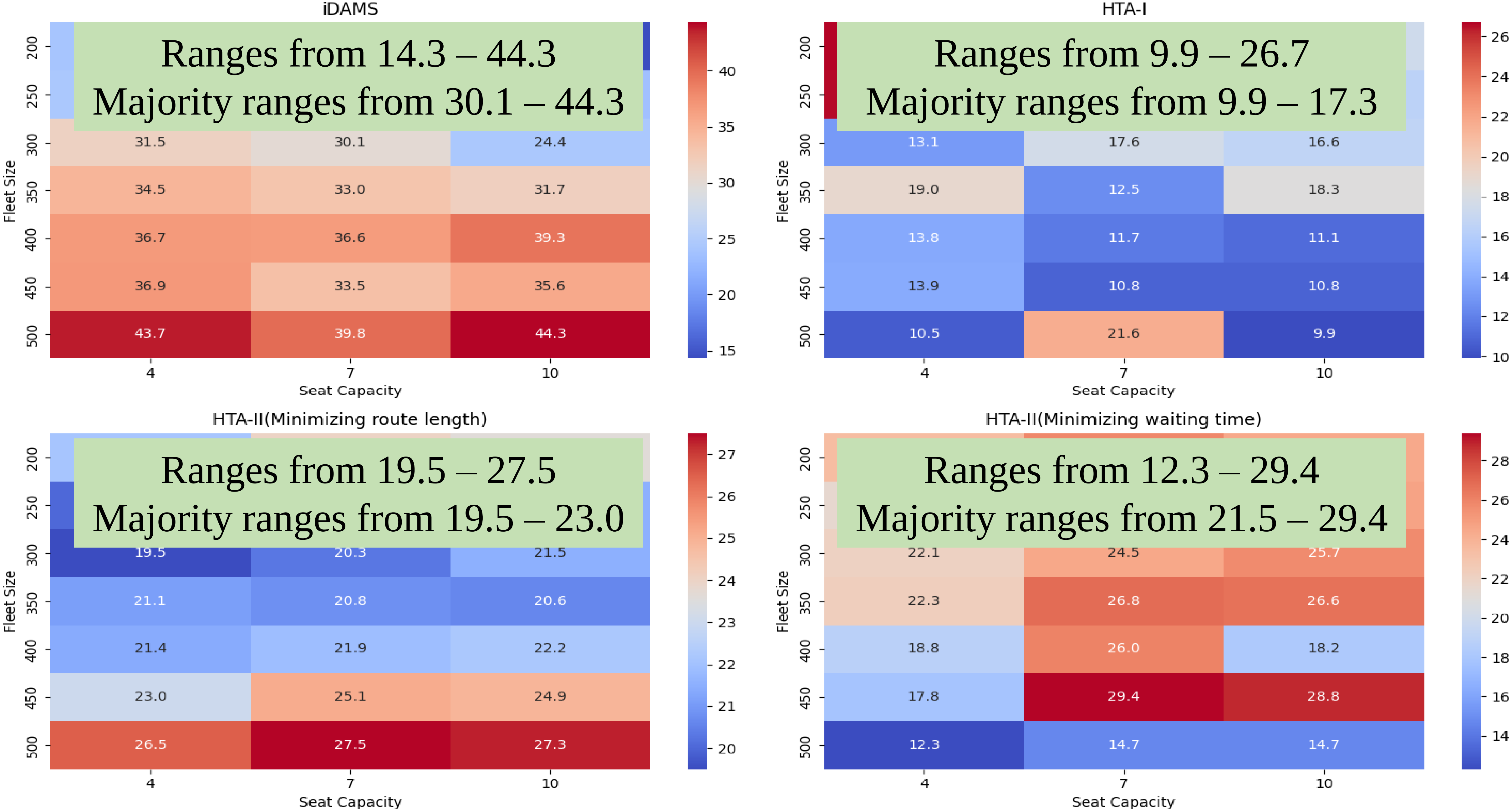


Figure: Standard deviation of waiting time

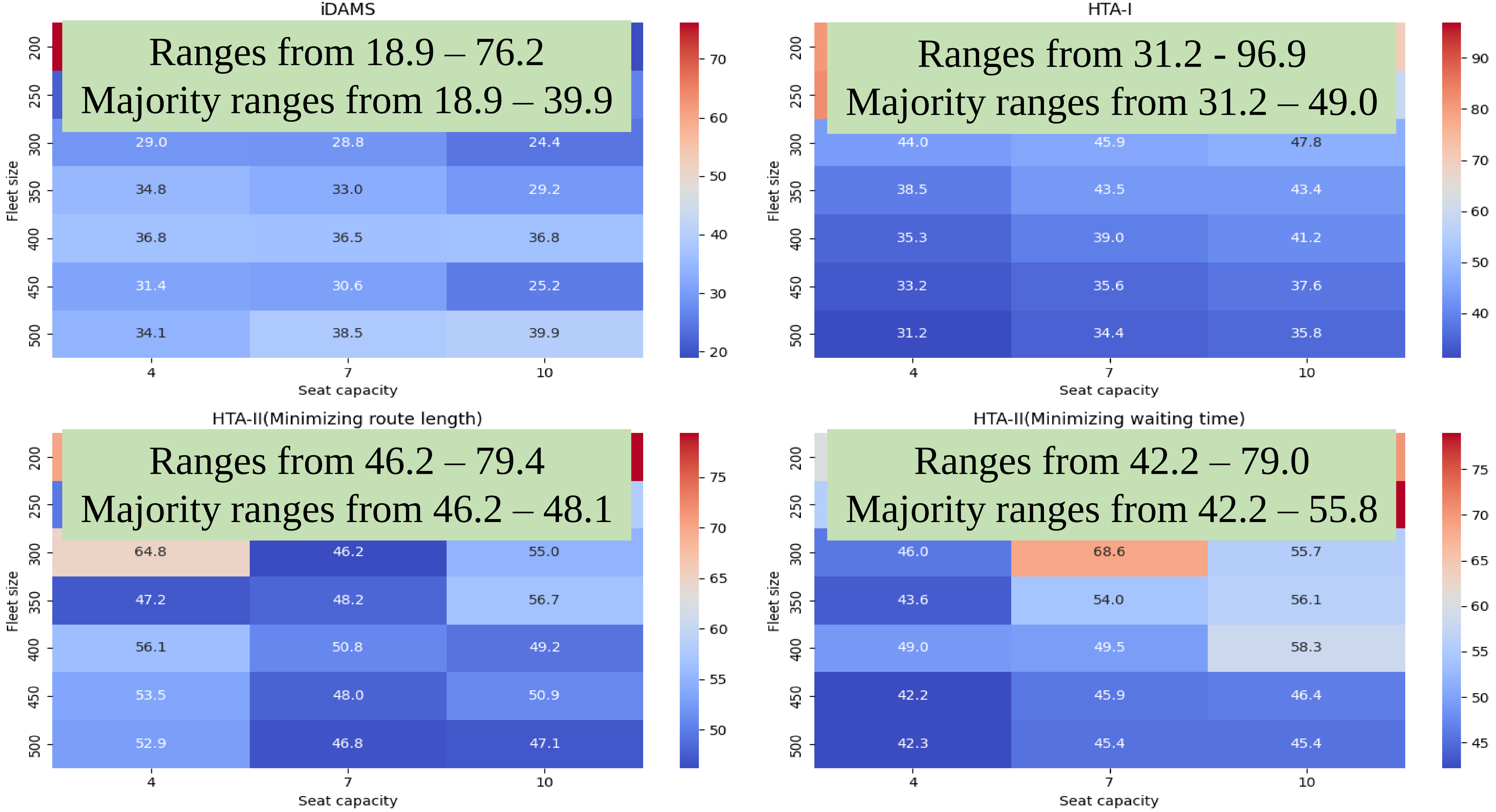


Figure: Standard deviation of delay time

COMPARISON: ALL ODT MODELS

	iDAMS (Anik et al. 2024)	HTA-I	HTA-II (Min. RL)	HTA-II (Min. WT)
Completed Trip (%)	97.39	**99.49**	93.11	98.44
Average indiv. wait (minutes)	8.83	**3.22**	12.95	4.19
Standard indiv. wait (minutes)	39.3	**10.8**	26.5	12.3
Average indiv. delay (minutes)	**9.51**	17.04	31.61	20.14
Standard indiv. delay (minutes)	36.8	**35.6**	52.9	42.3
VKT reduction (%)	12.12	**19.46**	2.70	4.93
Early reached (%)	19.65	**36.96**	9.65	31.28

COMPARISON: ALL ODT MODELS

	HTA-II (Min. RL)	HTA-II (Min. WT)
Completed Trip (%)	93.11	**98.44**
Average indiv. wait (minutes)	12.95	**4.19**
Standard indiv. wait (minutes)	26.5	**12.3**
Average indiv. delay (minutes)	31.61	**20.14**
Standard indiv. delay (minutes)	52.9	**42.3**
VKT reduction (%)	2.70	**4.93**
Early reached (%)	9.65	**31.28**

COMPARISON: ODT VS RIDESHARING

Model	Fleet size	Trip (%)
iDAMS (**Anik et al. 2024**)	10-400	97.39
HTA-I	7-450	99.50
HTA-II (Min. RL)	4-500	98.11
HTA-II (Min. RL)	4-500	98.71
ODHCRS (**Alonso-Mora et al. 2017**)	4-500	90.62

Best performance

Worst performance

FINDINGS

A central authority can serve the travel demand of a city

Choosing an appropriate objective function is a critical design decision

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CONCLUSION

- ✓ Objective function for **minimizing waiting time** generate more efficient solutions compared to minimizing route length
- ✓ **TSP-based formulations** outperformed VRPTW for online optimization
- ✓ ODT models significantly outperformed ridesharing model ODHCRS

FUTURE WORK

✓ Request will be filtered in terms of both individual **waiting time** and **delay time**

✓ Framework will be evaluated on **different benchmark datasets**

✓Systematic **parameter tuning** could further improve performance

✓ **Multi-objective optimization** frameworks could better navigate trade-offs between competing goals

ACKNOWLEDGEMENT

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Bangladesh University of Engineering and Technology

For providing dataset

DALTRAC

DALHOUSIE TRANSPORTATION COLLABORATORY

THANK YOU

APPENDIX

TERMINOLOGY: TIME COMPONENT

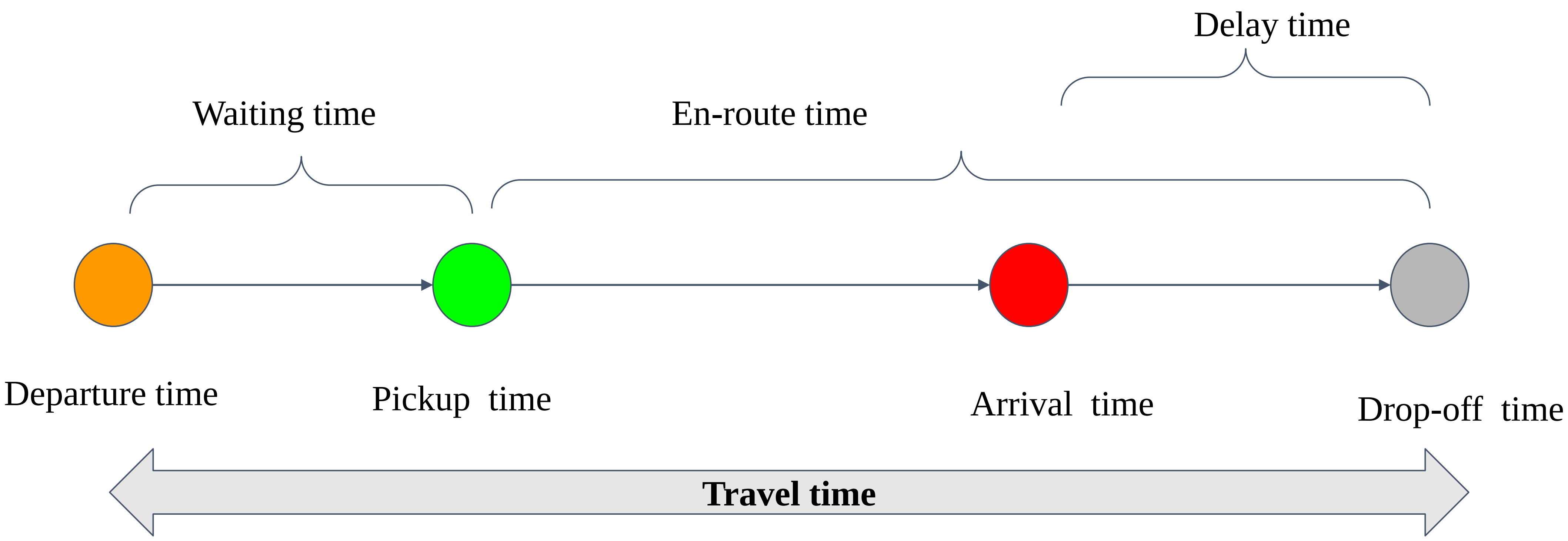


Figure: Timeline diagram indicating departure time, pickup time, waiting time, arrival time, en-route time, drop off time with the delay occurring situation

TERMINOLOGY: TIME COMPONENT

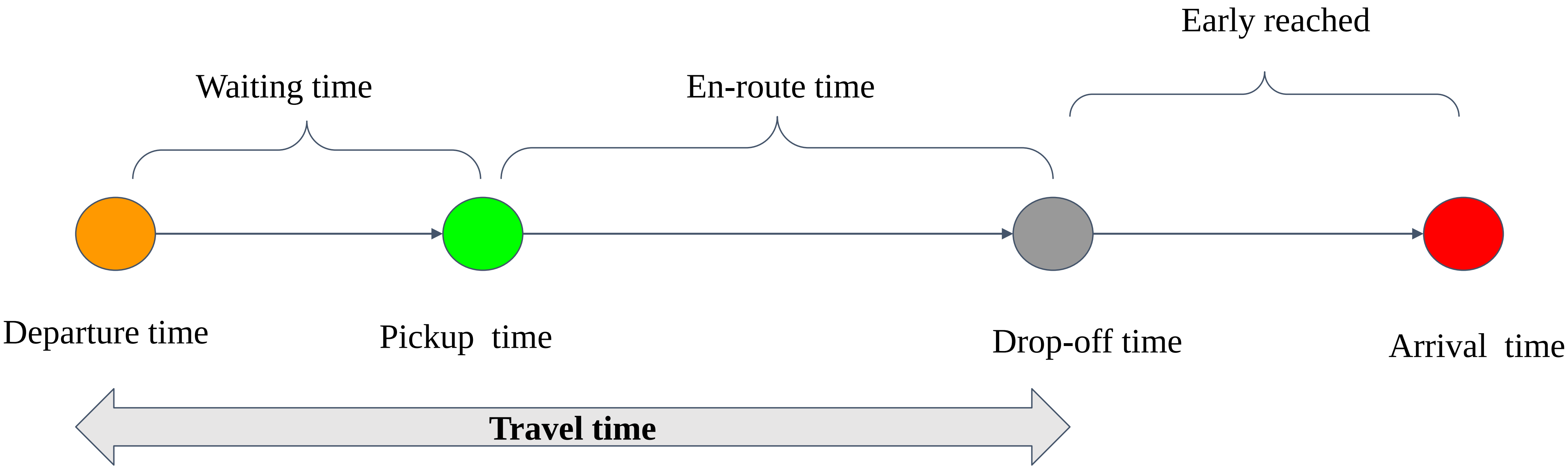


Figure: Timeline diagram indicating departure time, pickup time, waiting time, arrival time, en-route time, drop off time with the early reached condition

ALGORITHMS

TSP: Simulated Annealing

- 1-1 move: Swap two nodes
- Inner loop iteration: 10
- Max-iteration: 3
- Fitness/ objective : Min (Route length)

VRPTW: Hybrid Genetic Search(HGS)

- Parent selection: 2-ary tournament
- Crossover: Ordered crossover (OX)
- Fitness/ Objective: Min (penalty)
- Penalty/cost: Time window penalty (6)
- Stopping criteria: Check no improvement of best solution (2)

HTA-I: SIMULATED ANNEALING

$t \leftarrow$ temperature, initially a high number

100

$S \leftarrow$ some initial candidate solution

List of **unique** nodes

$Best \leftarrow S$

Route length

repeat

  $R \leftarrow$ Tweak(Copy(S))

1-1 movement

  if Quality(R) > Quality(S) or if a random number chosen from 0 to 1 $< e^{\frac{\text{Quality}(R)-\text{Quality}(S)}{t}}$ **then**

    $S \leftarrow R$

  Decrease t

temp = - temp*(.01)

  if Quality(S) > Quality($Best$) **then**

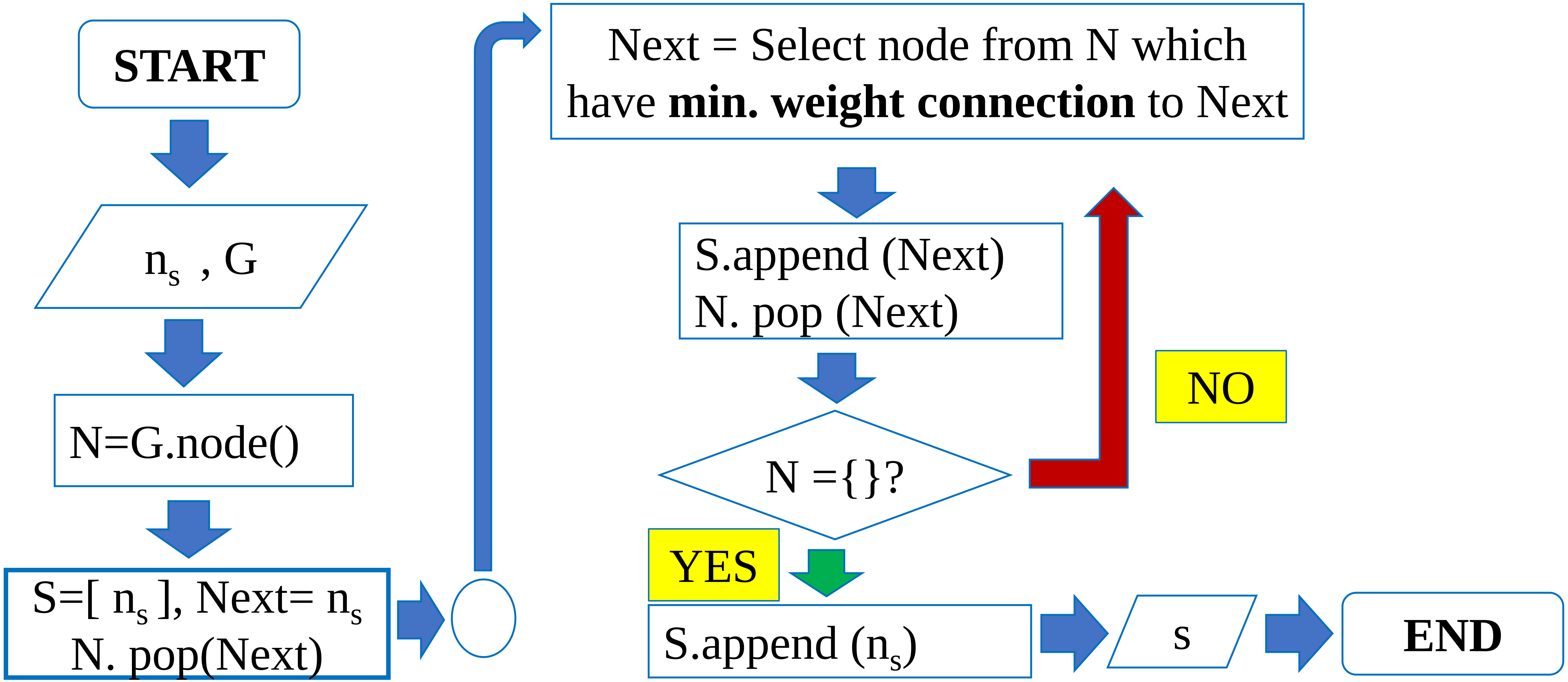
    $Best \leftarrow S$

until $Best$ is the ideal solution, we have run out of time, or $t \leq 0$

3 iterations

return $Best$

HTA-I: GREEDY INITIALIZATION OF SA



HTA-I: TWEAK (1-1 MOVEMENT)

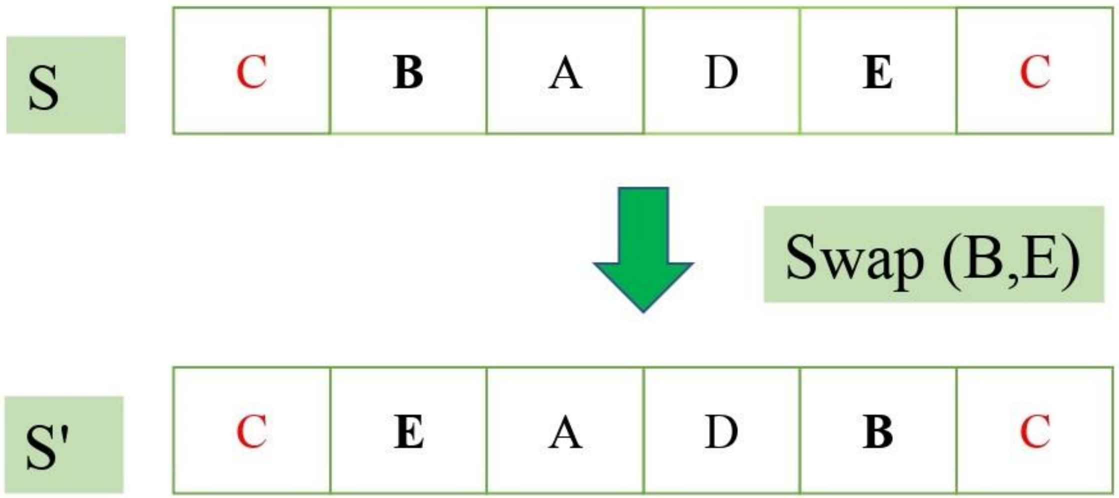


Figure: 1-1 movement

HTA-II: HYBRID GENETIC ALGORITHM

Hybrid Genetic Search(HGS)

Input: initial solutions s1,.....,sn

Output: the best-found solution s*

Set s*to the initial solution with the best objective value

repeat until the stopping criterion is met:

Select two parent solutions (sp1,sp2) from the population using k-ary tournament.

Apply crossover operator XO to generate an offspring solution **s0= XO(sp1,sp2)**

Improve the offspring using a search procedure *LS* to obtain **sc=LS(s0)**

Add the candidate solution to the population.

if sc has a better objective value than s*

s*<--sc

if population size exceeds maximum size:

Remove the solutions with the lowest fitness until the population is at a minimum size

return s*

Binary tournament selection

Ordered crossover

Local Search

40

25

HTA-II: TOURNAMENT SELECTION

$P \leftarrow$ population
$t \leftarrow$ $\boxed{\text{tournament size, } t \geq 1}$

t = 2

$Best \leftarrow$ individual picked at random from P with replacement
for i from 2 to t **do**
    $Next \leftarrow$ individual picked at random from P with replacement
    if Fitness($Next$) > Fitness($Best$) **then**
        $Best \leftarrow Next$
return $Best$

HTA-II: ORDERED CROSSOVER

Two parents P1 and P2

P1	1	2	3	4	5	6	7	8	9
P2	9	3	7	8	2	6	5	1	4

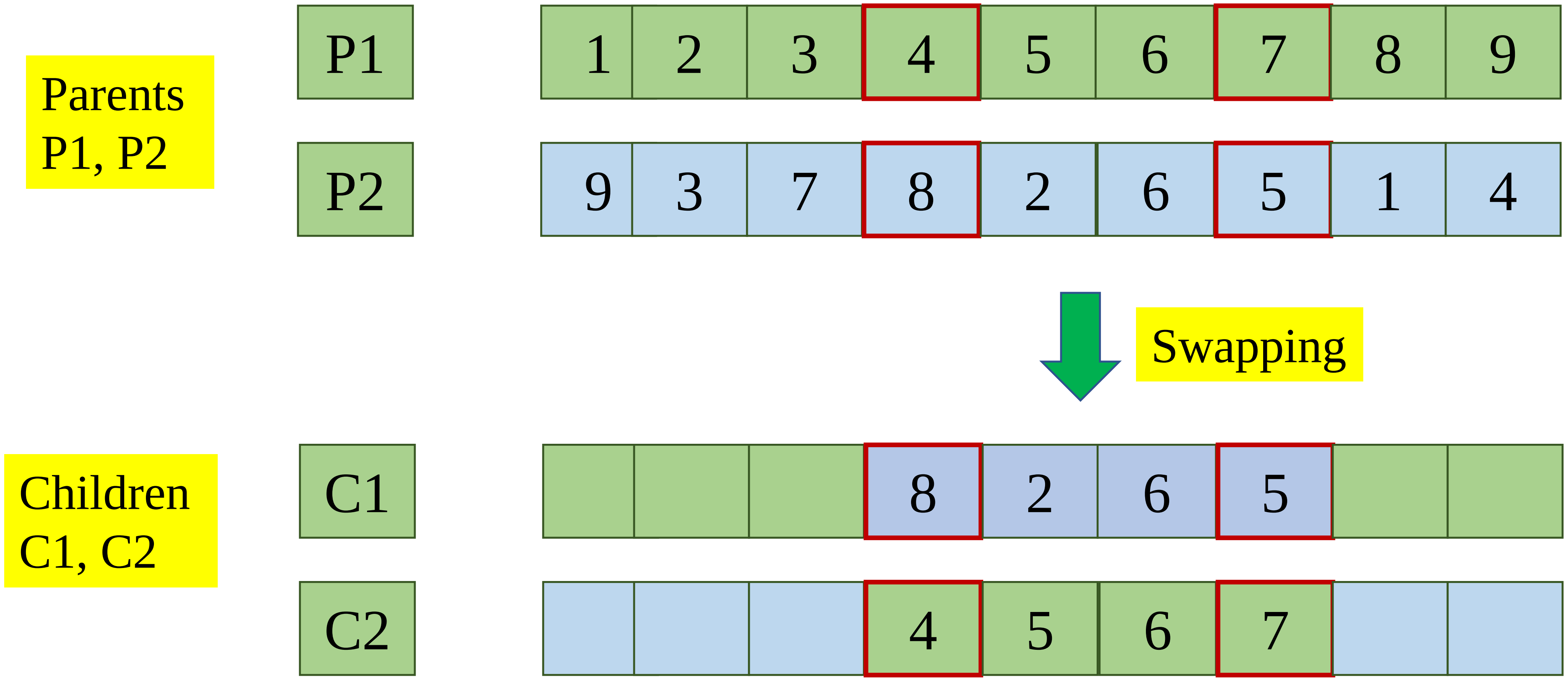
HTA-II: ORDERED CROSSOVER

Two parents P1 and P2

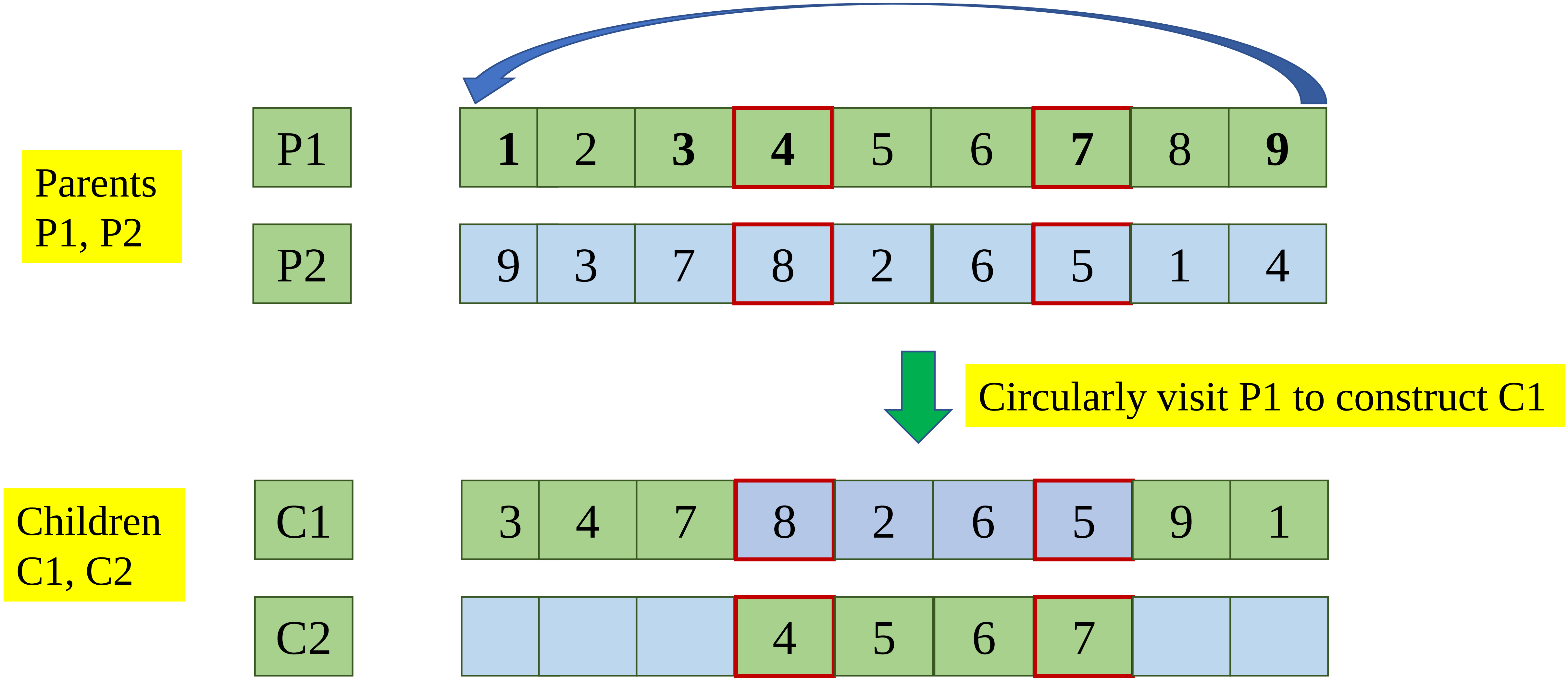
P1	1	2	3	4	5	6	7	8	9
P2	9	3	7	8	2	6	5	1	4

Select two random points

HTA-II: ORDERED CROSSOVER



HTA-II: ORDERED CROSSOVER



HTA-II: ORDERED CROSSOVER

Parents
P1, P2

P1	1	2	3	4	5	6	7	8	9
P2	**9**	**3**	7	**8**	**2**	6	5	**1**	4

Circularly visit P2 to construct C2

Children
C1, C2

C1	3	4	7	8	2	6	5	9	1
C2	3	8	2	4	5	6	7	1	9

HTA-II: TERMINATION CONDITION

Max iteration

Max runtime

No improvement

- Infeasible solution
- Run time is high

PERFORMANCE METRICS: SUMMARY

Performance metrics	Comment	Goodness
Trip completion (%)	Overall system completeness	High is good
Average indiv. wait	Travelers cost	Low is good
Average indiv. delay	Travelers cost	Low is good
VKT reduction (%)	Vehicle cost	High is good
Early reached	Travelers saving	High is good

COMPARARISON: iDAMS VS HTA-I

	iDAMS (Anik et.al, 2024)	HTA-I
Seat capacity- Fleet size	10-400	7-450
Completed Trip (%)	97.39	**99.49**
Average indiv. Wait	8.83	**3.22**
Average indiv. delay	**9.51**	17.04
VKT reduction (%)	12.12	**19.46**
Early reached (%)	19.65	**36.96**

COMPARISON: HTA-II (Min. RL) VS HTA-II (Min. WT)

	HTA-II (Min. RL)	HTA-II (Min. WT)
Seat capacity- Fleet size	4-500	4-500
Completed Trip (%)	93.11	**98.44**
Average indiv. Wait	12.95	**4.19**
Average indiv. delay	31.61	**20.14**
VKT reduction (%)	2.70	**4.93**
Early reached (%)	9.65	**31.28**

COMPARISON: iDAMS VS HTA-II (Min. RL)

	iDAMS (Anik et.al, 2024)	HTA-II (Min. RL)
Seat capacity- Fleet size	10-400	4-500
Completed Trip (%)	**97.39**	93.11
Average indiv. Wait	**8.83**	12.95
Average indiv. delay	**9.51**	31.61
VKT reduction (%)	**12.12**	2.70
Early reached (%)	**19.65**	9.65

COMPARISON: HTA-I VS HTA-II (MIN. WT)

	HTA-I	HTA-II (MIN. WT)
Seat capacity- Fleet size	7-450	4-500
Completed Trip (%)	**99.49**	98.44
Average indiv. Wait	**3.22**	4.19
Average indiv. delay	**17.04**	20.14
VKT reduction (%)	**19.46**	4.93
Early reached (%)	**36.96**	31.28

REFERENCES

Zidi, I., Mesghouni, K., Zidi, K., & Ghedira, K. (2012). A multi-objective simulated annealing for the multi-criteria dial a ride problem. *Engineering Applications of Artificial Intelligence*, *25*(6), 1121-1131.

Huang, A., Dou, Z., Qi, L., & Wang, L. (2020). Flexible route optimization for demand-responsive public transit service. *Journal of Transportation Engineering, Part A: Systems*, *146*(12), 04020132.

Masmoudi, M. A., Braekers, K., Masmoudi, M., & Dammak, A. (2017). A hybrid genetic algorithm for the heterogeneous dial-a-ride problem. *Computers & operations research*, *81*, 1-13.

REFERENCES

Huang, D., Gu, Y., Wang, S., Liu, Z., & Zhang, W. (2020). A two-phase optimization model for the demand-responsive customized bus network design. *Transportation Research Part C: Emerging Technologies*, *111*, 1-21.

Cao, Y., & Wang, J. (2017). An optimization method of passenger assignment for customized bus. *Mathematical Problems in Engineering*, *2017*(1), 7914753.

Anik, M. A. H., Nayeem, M. A., & Habib, M. A. (2024). Development of a dynamic traffic microsimulator for on-demand transit operations within an integrated modelling system. *Transportation Planning and Technology*, 1-27.

REFERENCES

Alonso-Mora, J., Samaranayake, S., Wallar, A., Frazzoli, E., & Rus, D. (2017). On-demand high-capacity ride-sharing via dynamic trip-vehicle assignment. *Proceedings of the National Academy of Sciences*, *114*(3), 462-467.

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